Automation in Construction 190 (2026) 107076

Contents lists available at ScienceDirect

Automation in Construction

journal homepage: www.elsevier.com/locate/autcon

Digital twin-enabled synchronous jacking control of an aerial building machine using a scale model

Limao Zhang [a,b], Xupu Zhang [a], Ke Chen [a,b], Jiaqi Wang [a,*], Lieyun Ding [a,b]

[a] School of Civil and Hydraulic Engineering, Huazhong University of Science and Technology, 1037 Luoyu Road, Hongshan District, Wuhan, Hubei 430074, China
[b] National Center of Technology Innovation for Digital Construction, Huazhong University of Science and Technology, 1037 Luoyu Road, Hongshan District, Wuhan, Hubei 430074, China

ARTICLE INFO

Keywords:
ABM
Scale model
Synchronous control
Transfer learning
Deep learning

ABSTRACT

The aerial building machine (ABM) provides an integrated solution for high-rise construction through its ability to hoist construction platforms, materials, and equipment to elevations of several hundred meters while maintaining high operational adaptability. The jacking process of the ABM depends on coordinated extension and retraction of multiple cylinders. However, high-altitude wind loads and non-uniform stacking loads on the construction platform introduce significant uncertainties that may compromise structural safety. This paper presents the development of an intelligent synchronous control system based on a scaled experimental platform, employing a deep neural network for dynamic error compensation, together with a digital twin framework to improve operational efficiency and human–machine collaboration. Reinforcement learning tuned control parameters, and transfer learning migrated DNN-PID controller to the scale model. Jacking tests showed 0.407 mm synchronization error, a maximum error below 0.46 mm across ten conditions, and a 29% maximum synchronization error reduction with fewer iterations.

1. Introduction

The number and average height of super tall buildings in the world have continued to increase steadily in recent years, showing a sustained growth trend. As of February 2025, China has a total of 3465 super high-rise buildings over 150 m in height and 122 super high-rise buildings over 300 m in height [1], ranking first globally in both categories. Super tall buildings inherently have the characteristics of maximizing land resource utilization, and can meet the demands of functional use and urban aesthetics [2]. However, compared with low-rise buildings, the construction of high-rise buildings has the characteristics of high construction difficulty, high safety risk, and complex processes [3]. Most notably, the deformation, collapse, and overturning of the scaffolding system have resulted in severe production safety accidents and substantial losses [4]. Additionally, the traditional scaffolding construction mode cannot efficiently integrate material transportation, concrete pouring, and steel reinforcement trapping. Construction equipment such as tower cranes and fabric machines is not able to quickly climb with the construction of the building core, leading to low construction efficiency and difficulties in cost control during the construction cycle of high-rise buildings. To solve the problems of high-altitude operations, the aerial building machine (ABM) was developed [5].

ABM is a self-jacking scaffolding system with a steel platform, which has a complete sensing and control system. Its main subsystems include the power system, support system, and formwork system [6], which is characterized by high load capacity, strong adaptability, easy operation, intelligent control, and comfortable environment [7–9]. In order to achieve its switching under various working conditions and complete self-climbing for the construction of high-rise buildings, its support and power systems need to smoothly complete the jacking of the steel platform [10]. The ABM integrates multiple functions, and its complex structure and diverse working conditions impose stringent requirements on the stability of the power system and control equipment. Any instance of asynchronous jacking necessitates an immediate construction halt for troubleshooting and can lead to severe consequences such as structural damage [10,11]. Therefore, millimeter-level synchronization accuracy is important in the control process, and high-performance control algorithms are required to drive the synchronous jacking process of multiple oil cylinders and effectively correct asynchronous climbing in real time, thereby ensuring the precise levelness of the steel platform.

* Corresponding author.
E-mail addresses: zlm@hust.edu.cn (L. Zhang), xpzhang@hust.edu.cn (X. Zhang), chenkecm@hust.edu.cn (K. Chen), jiaqi_wang@hust.edu.cn (J. Wang), dly@hust.edu.cn (L. Ding).

https://doi.org/10.1016/j.autcon.2026.107076
Received 15 September 2025; Received in revised form 28 May 2026; Accepted 1 June 2026
Available online 15 June 2026
0926-5805/© 2026 Published by Elsevier B.V.

These engineering challenges place strict requirements on the reliability and synchronization performance of the jacking control system, motivating the development of more advanced synchronous control strategies for ABM platforms [12].

Traditional synchronous jacking control methods used in ABM systems still exhibit limitations such as insufficient control accuracy, slow response speed, and insufficient real-time monitoring feedback under complex working conditions, and in some cases still rely on manual observation of platform levelness for safety management. In current research and engineering practice, multi-cylinder synchronous control methods enhance their control performance mainly through improvements in target tracking accuracy and advanced coupling strategies [13]. These synchronous jacking control methods are primarily based on parallel [14] or master-slave [15] approaches, which often exhibit poor anti-interference capability [16]. Coupling structures [17,18] have improved this aspect. In applications where operating conditions are relatively stable and system parameters can be accurately modeled, closed-loop control methods have evolved toward adaptive, robust, and intelligent approaches. Representative examples include fuzzy adaptive control [19], sliding mode control [20], neural-network-based controllers such as DNN-PID [21], generalized predictive control (GPC) [22], as well as synchronization strategies based on deviation-coupling structures [23] and FPGA-based digital synchronization [24]. Although existing methods have improved multi-cylinder synchronization accuracy, they typically rely on accurate system modeling and relatively stable operating conditions. In practical ABM jacking scenarios, varying loads and inconsistencies in actuator hydraulic circuits make accurate system parameterization difficult, which limits the applicability of model-based control approaches. Precisely because of these modeling challenges, engineering practice often continues to rely on traditional PID-based synchronization strategies, which may become insufficient under complex operating conditions [25].

The application of the jacking control method in actual operations is the most effective way to validate its effectiveness. In practice, field observations provide valuable data for safety management and iterative improvement of control strategies. However, setting up multiple operating conditions and conducting comparative experiments with different algorithms is difficult to achieve in actual construction processes. Due to the high risks involved in high-rise building construction and the high costs of experiments, algorithms are difficult to directly deploy in ABM equipment, and control parameters and attitude feedback during asynchronous jacking are also challenging to obtain. To address these issues, current mainstream solutions include combined simulation of hydraulic and control circuits [12]. Simulation systems generally struggle to accurately replicate real-world operational states due to factors such as the inability to account for random fluctuations in hardware and circuits, as well as numerous constraints. Multi-domain research has proposed using scale models to simulate system motion states [26]. This paper employs a scale model to reconstruct the main structure and power system of the ABM and applies it to a jacking test execution platform. This provides a pathway for the research and experimental validation of synchronous control methods.

Although the methods mentioned can achieve acceptable performance, they often involve high training or tuning costs and slow convergence, especially under varying operating conditions. These limit their practicality in real ABM applications. Moreover, closed-loop digital twin control frameworks validated through physical experimentation in ABM systems remain largely absent. To address this issue, transfer learning is introduced to bridge simulation-based optimization and physical deployment. By leveraging prior control experience from simulation, the controller can be initialized in the physical system with reduced manual tuning, improving convergence efficiency and robustness under asymmetric and dynamic loads. To support systematic validation and management of control strategies, a digital twin framework is introduced as a system-level support layer. Through bidirectional interaction with the scaled model, it enables structured testing, real-time monitoring, and traceable evaluation. This design is intended to support future deployment and system integration, rather than implying direct applicability to full-scale ABM systems at this stage.

In summary, this paper addresses the following three key issues: (1) How to design the ABM scale model platform for synchronous jacking validation; (2) How to develop an online-learning adaptive controller capable of maintaining multi-cylinder synchronization under non-stationary load conditions; (3) How to drive the intelligent control and management of synchronous jacking control with digital twins, and provide a structured experimental basis for future investigation of control strategies in ABM systems. To solve the above problems, this paper proposes a synchronous control framework based on the scale model, transfer learning, and digital twins, which realizes adaptive intelligent synchronous control of multiple motors under uneven loads. The contributions of this research lie in: (a) A scaled ABM experimental platform is developed to enable safe and repeatable validation of synchronous jacking control strategies under diverse loading conditions; (b) A transfer learning–based DNN-PID control strategy is proposed to bridge simulation and physical deployment, improving convergence efficiency and synchronization performance under non-stationary conditions; (c) A digital twin–supported framework is established to coordinate control execution, data logging, and experimental management, providing a structured basis for future investigation and potential deployment in real ABM systems.

The three main objectives of this paper are listed below: (1) Scaled-Model Design: A scaled-down experimental platform was designed and developed, integrating a sensor suite and an embedded control system to address the practical challenge of validating algorithmic feasibility under real-world conditions, where direct experimentation is often infeasible. (2) Intelligent Synchronization Control: A transfer learning–driven DNN-PID control strategy is proposed, leveraging prior knowledge acquired from simulation to achieve high-precision motor synchronization under asymmetric load conditions in physical environments of scale model. (3) Digital Twin–Enabled Management and Control System: A full-lifecycle digital twin framework is established, enabling integrated closed-loop management and control through data mapping and visualization, thereby laying the foundation for the deployment of the proposed methods in real-world engineering systems. Together, these three components form a closed-loop research framework. Transfer learning bridges simulation-based controller development and physical experimentation on the scale model, while the digital twin enables system-level coordination and monitoring of the experimental process, thereby linking algorithm adaptation, physical validation, and system-level management within a unified environment.

The remaining research contents of this study are organized as follows. Section 2 provides the related studies on synchronous control. Section 3 introduces the details of the intelligent control algorithm. Next, in Section 4, the intelligent control algorithm is employed in a scale model of ABM to validate its effectiveness and practicability. Section 5 demonstrates the superiority and key links of intelligent algorithms. In the last section, the conclusions and future works are presented.

2. Related studies on synchronous jacking control

This paper focuses on the synchronous control of the ABM jacking process under varying construction conditions. In this scenario, control performance is jointly influenced by the adaptability of the control algorithm, the sim-to-real transferability of the control strategy, and the integration level and systemic completeness of the control framework. Accordingly, the integration of intelligent algorithms, simulation-to-scale transfer learning, and digital twin–driven control constitutes a comprehensive control framework that addresses complex challenges from complementary perspectives. Numerous studies have investigated the application of intelligent algorithms, transfer learning, and digital twin technology in engineering. The relevant research can be

summarized as follows:

(1) *Intelligent algorithms in synchronous control*. Intelligent synchronization control algorithms are providing robust solutions for networked systems under topology uncertainty, random perturbations, and resource-constrained scenarios through adaptive, event-triggered, neural network, and hybrid strategies. Wang et al. [27] use the artificial bee colony (ABC) algorithm to optimize the Relative coupling control (RCC) structure for a multimotor synchronous control system and improve its coupling performance. Xie et al. [28] proposed a generalized model of a dual parallel motion system to solve the problem of large differences in control objectives and mechanical structures in parallel control systems. Shouran et al. [29] proposed a controller combining fuzzy PIDF with fractional order proportional differentiation (FOPD) to minimize the ITAE objective function by particle swarm optimization (PSO) and sand cat swarm optimization (SCSO) algorithms to improve the disturbance resistance by 40%. Xu et al. [30] proposed a multilevel fuzzy-PID control strategy to solve the problems of synchronization speed deviation and positioning error caused by time-varying and nonlinearities in the jacking synchronization control system. Chen et al. [31] proposed a recursive Chebyshev fuzzy neural network (RCFNN)–based current angle controller for synchronous reluctance motors, capable of searching for the maximum power factor point and improving efficiency across varying operating conditions. Their practical implementation relies on iterative offline debugging and precise system modeling. In certain scenarios, they even sacrifice other performance metrics to optimize for specific task objectives. In ABM jacking applications, challenges persist in directly deploying such algorithms due to continuously varying load distributions, fluctuating actuator performance across construction phases, and extremely limited on-site debugging opportunities.

(2) *Transfer learning of control strategies*. Transfer learning (TL) aims to enhance the performance of target learners in the target domain by transferring knowledge from different but related source domains, thereby addressing challenges such as data scarcity and distribution shifts in engineering applications [32]. Kuang et al. [33] proposed an unbalanced adversarial transfer learning network for industrial fault diagnosis, addressing data scarcity under fault conditions and extending the applicability of transfer learning. In additive manufacturing, Pashmforoush et al. [34] used TL to predict mechanical properties of selective-laser-melted materials by transferring simulated data to physical experiments, slashing data needs by 50% with $<5\%$ error, and to guide the optimization of control strategies for this process. Wang et al. [35] transferred the knowledge learned by the transient stability assessor (TSA) established by the graph convolutional neural network (GCNN) to the entropy-enhanced twin delay deep deterministic policy gradient (DRL) model to improve the learning efficiency and performance of transient stability preventive control (TSPC). Yin et al. [36] proposed a deep transfer reinforcement learning method based on QMIX that accelerates convergence through policy transfer and enables multi-UAV task allocation. Gadiraju et al. [37] used a method that utilizes the transferability of CNN networks to obtain pre-trained weights from a crack image dataset, thereby completing crack identification and detection using drones and improving the performance of crack identification and localization. Existing research on transfer learning primarily focuses on perception, diagnosis, or high-level decision-making tasks, while its functional role as a controller remains underdeveloped. Ideally, such methods should be integrated with system dynamics to serve as an integral part of the closed-loop control. Consequently, its applicability to real-time synchronous control of multi-actuator systems remains limited.

(3) *Digital twin-driven control*. Driven by data and models, digital twin (DT) promotes interaction and integration between the physical world and the information world [38] and can perform functions such as monitoring, simulation, prediction, and optimization [39]. Some scholars have pointed out that DT-enhanced parallel computing can accelerate the execution of real-time optimization algorithms in closed-loop control systems, guiding the application of digital twin methods in the field of control [40]. DT architecture has practical applications in the research of engineering equipment such as tunnel boring machines (TBM) and tower cranes. Jiang et al. [26] developed a scale model–based digital twin framework to simulate tower crane jacking behavior and analyze its dynamic effects, providing experimental insights for monitoring unsafe jacking operations. Zhang et al. [41] established a digital twin model of a TBM integrated with deep learning and online multi-objective optimization to determine optimal control parameters, and validated its effectiveness using real metro construction data. Existing digital twin applications play different roles across construction equipment. For tower cranes, control remains largely operator-driven, and digital twins mainly support monitoring and safety assistance rather than real-time control. ABM jacking, by contrast, requires a high degree of automation, where operators specify target states and synchronous control is executed by the control system itself. Although this makes ABM superficially similar to TBM systems, TBMs typically rely on dense sensory information for environmental perception and decision support. ABM jacking must instead maintain reliable multi-actuator synchronization under uneven and time-varying loads with limited environmental sensing, and load information is rarely directly incorporated into the control loop. This difference limits the direct transferability of existing DT frameworks and highlights the need for a control-oriented digital twin tailored to ABM systems.

The research in the field of multi-cylinder synchronous control technology has developed a relatively mature technical route from theory to implementation, among which the PID control strategy optimized by intelligent optimization algorithms and deep neural networks has shown strong performance and stability. However, most existing studies implicitly assume stable operating conditions, accurately modelable system parameters, and high actuator consistency, which limits their direct applicability in complex construction scenarios. As a large-scale civil engineering equipment that has emerged in recent years, the control and safety management system of the ABM in engineering practice is still in the process of construction and improvement. The safety of the jacking process of some ABM used in super high-rise building construction projects still relies on ground observation and manual control by operators. At present, systematic research focusing on digital twin-driven jacking modes and safety risk mitigation for ABM systems remains limited. Furthermore, the data links in existing Digital Twin (DT) research remain insufficiently defined. Most studies are either based on simulations or rely on a unidirectional data flow from physical engineering equipment to DT systems. Under the constraints of real-world conditions, there is a lack of a physical entity capable of establishing bidirectional data links within the DT closed loop. Consequently, research using scaled models to fulfill this role and provide closed-loop feedback remains limited. By integrating transfer learning for control adaptation, scale-model-based validation, and digital twin–based system-level coordination, this paper aims to bridge the gap between algorithmic advances and practical ABM jacking control systems, thereby promoting the automation and intelligent development of large-scale construction equipment.

3. Methodology

This paper investigates synchronous jacking control of ABM using an experimental scale model. For the asynchronous jacking problems that may occur in the jacking process, a set of monitoring, analysis, control, and early warning programs is designed to reduce the safety risk and extend the service life of the equipment. Firstly, key structural components are replicated according to similarity principles to complete the design, manufacture, and assembly of the scale model together with the corresponding digital model, and then the multi-objective optimization reinforcement learning algorithm is adopted to optimize the dynamic control parameters of the scaled model, together with a DNN-PID–based synchronous control strategy for the multi-motor system. Subsequently, the single-motor and multi-motor tracking simulation models are constructed in MATLAB/Simulink. The DNN is trained in the simulation and adapted to the physical environment by transfer learning. Finally, the synchronization performance is evaluated through jacking experiments using maximum steady-state error (MSSE), maximum synchronization error (MSE), average synchronization error (ASE), and other indexes, and a twin system integrating data interaction, safety warning, and remote control is constructed. The framework is shown in Fig. 1.

3.1. System architecture and scale model

This section introduces the system architecture of the proposed experimental platform for synchronous jacking control of the ABM. The system integrates a physical jacking platform, an embedded control system, and a digital twin interface to support control execution and data interaction. The design of the platform is described from three aspects: the overall system architecture, the scaled ABM model and the modeling of the jacking scenario, and the embedded control system.

3.1.1. Physical system configuration

The study of the synchronous jacking control algorithm was conducted on a self-developed scaled experimental platform. From the perspective of system architecture, the platform consists of three parts: a physical jacking system, an embedded hardware control system, and a data communication interface reserved for digital twin integration. The physical jacking system includes the structural components of the scaled model, electric linear actuators used to execute the jacking motion, and sensors for state acquisition. The actuators are responsible for realizing the lifting motion of the steel platform, while the sensors are used to obtain the operating state information of each actuator. The embedded control system is built on an STM32 series microcontroller. It is used to perform sensor signal acquisition, control computation, and actuator driving. Each actuator motor is equipped with an encoder to provide displacement and speed feedback signals, which are collected and processed through the timer interfaces of the controller.

During each control cycle, the controller drives the motor driver boards through PWM signals, thereby regulating the motion state of the actuators and enabling coordinated jacking control among the four actuators. In addition, the system is configured with a serial communication interface to enable bidirectional data exchange between the embedded controller and the upper-level service system. Through this interface, the upper-level system can transmit control targets and commands to the embedded controller, while the controller returns the motor operating states and displacement data in real time, providing support for system monitoring, log recording, and visualization in the digital twin platform. Through the coordinated operation of the physical system, the embedded control system, and the communication interface, an experimental platform architecture is established for synchronous jacking control experiments and data interaction. To clarify the physical process represented by the experimental platform, the jacking mechanism of the ABM system is briefly introduced below. The scale model

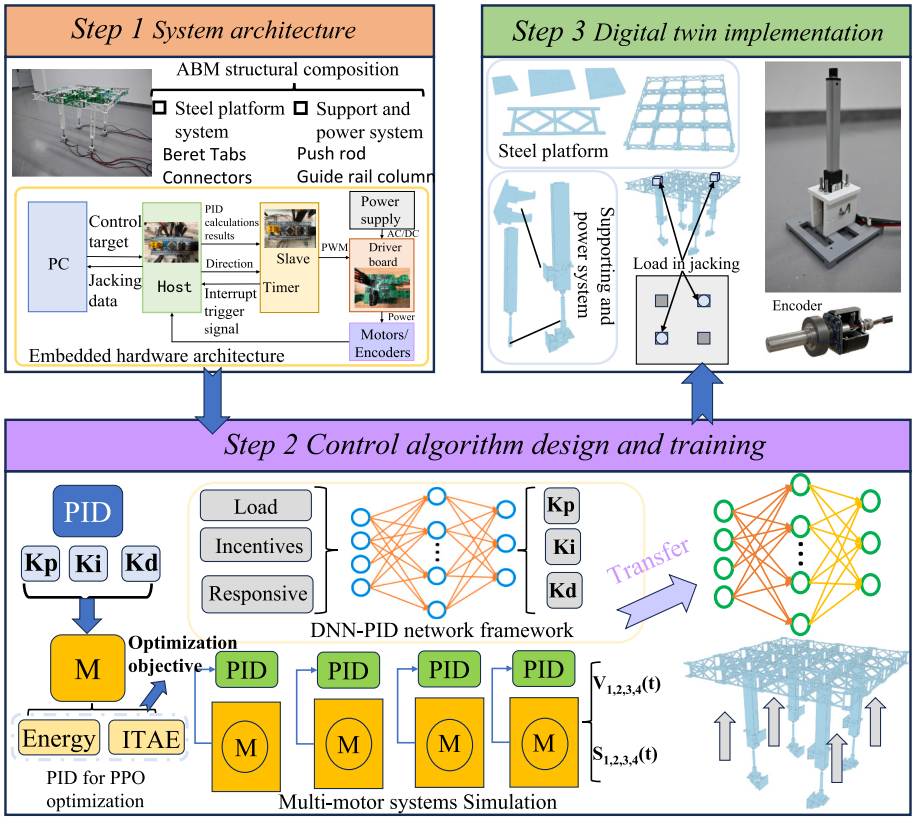


Fig. 1. Framework of ABM synchronized jacking control.

used in this study reconstructs the key structural components and motion stages of the ABM supporting and power system.

The supporting and power system of ABM consists primarily of steel columns, supporting equipment, cylinders, and supporting frames. Specifically, the steel columns are held in place with a steel platform above. The supporting frame is connected to the steel column by hydraulic cylinders. The steel platform is extended and retracted synchronously by several hydraulic cylinders, thus enabling the construction of high-rise buildings. The workflow of the protrusion operation begins with the cylinders extending to raise the steel columns and the steel platform, which are then secured by the equipment supporting the new position. Subsequently, the cylinder retracts and moves to a higher position on the support frame. This sequence of extension and retraction of the cylinder is repeated continuously throughout the jacking process. Under normal circumstances, the hydraulic cylinder is extended and retracted multiple times to raise the platform to a height corresponding to one floor of the building. The complete jacking process generally consists of five jacking stages of 700 mm, 700 mm, 700 mm, 800 mm, and 800 mm. Fig. 2 illustrates the complete jacking process, from the first extension of the hydraulic cylinder to the completion of the climbing frame's retrieval. In this study, the climbing phase is omitted since it does not involve synchronization, automation, or structural safety concerns. The scale model is constructed based on the extension and retraction stages that constitute the core synchronous control task.

3.1.2. ABM scale model

The scale model represents the physical entity dimension of the DT five-dimensional model [42] and serves as the experimental platform for synchronous jacking control research, as well as for verifying the reliability and robustness of the proposed control strategies. Similarity theory guides the development of the scale model in multiple aspects, such as size and shape structure [43]. In the ABM construction system, the components that most significantly influence the motion state and mechanical characteristics of the main structure are the support system, power system, and steel platform. Therefore, the establishment of the scale model focuses on these key structural components, with a similarity coefficient of 1:25, which is reflected in the dimensional scaling of the steel platform and the support system. This paper focuses on the levelness of the structure during the jacking process. The steel platform mainly serves to bear the load, and its static characteristics have limited influence on jacking levelness, therefore, PLA material is used for its fabrication. The power system of the actual ABM consists of 12 hydraulic cylinders, which are simplified to 4 motor-driven actuators in the scale model to achieve the jacking of the steel platform. As shown in Fig. 3(a), numerous components were constructed for assembling the scale model, in which modules ①-③ are the connectors, module ④ is the bailey truss, module ⑤ is the supporting column, module ⑥ is the motor, and module ⑦ is the steel platform. Fig. 3(b) shows the ABM scale model used for the jacking test.

3.2. Control algorithm design and training

This section describes the design and training process of the proposed synchronous jacking control method. The control framework is built on an improved deviation coupling structure to achieve coordinated control among multiple actuators. In the virtual environment, the PPO algorithm is first employed to perform multi-objective parameter optimization, obtaining a set of control parameters used to initialize the DNN. Subsequently, the DNN-PID controller is trained in the simulation environment and then transferred to the physical scaled experimental platform for validation. Multiple synchronization performance metrics are used to evaluate the control performance in the physical experiments.

3.2.1. PID parameter optimization based on PPO

PID parameter tuning is a type of multi-objective optimization task that requires finding a balanced optimal solution among multiple indicators. Reinforcement learning based on the PPO algorithm is used to

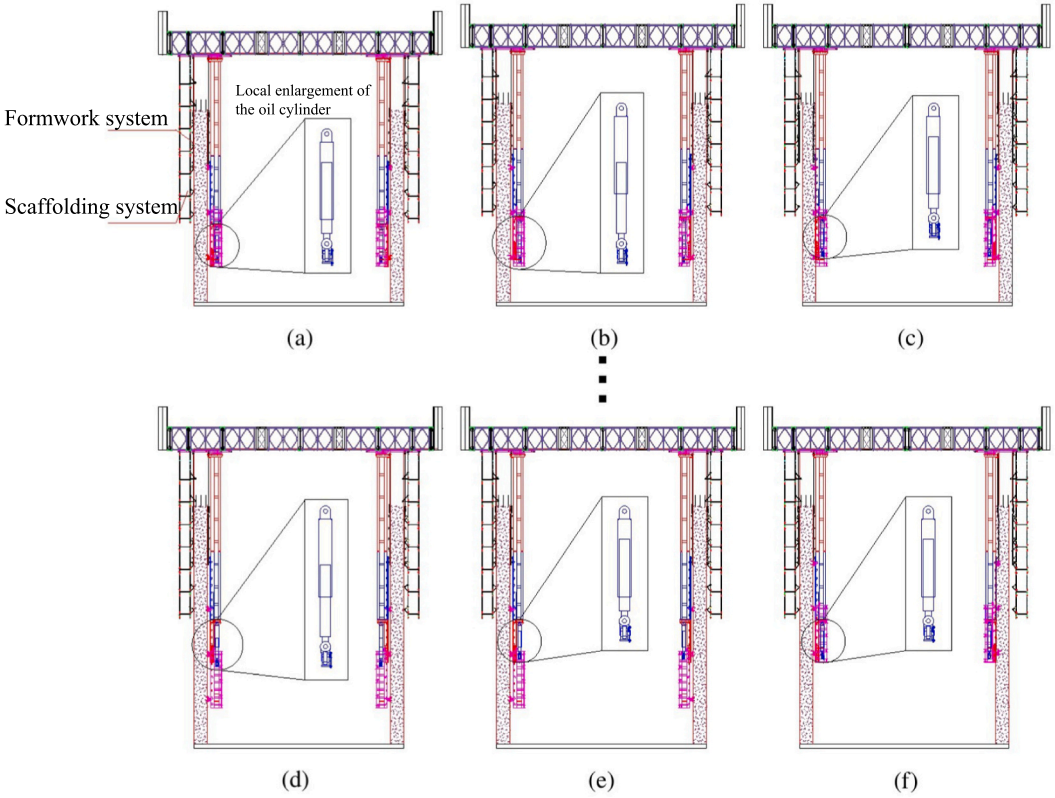


Fig. 2. ABM jacking process: (a) Ready, (b) First jacking, (c) First self-climbing, (d) Last jacking, (e) Last self-climbing, and (f) Climbing boots recycling.

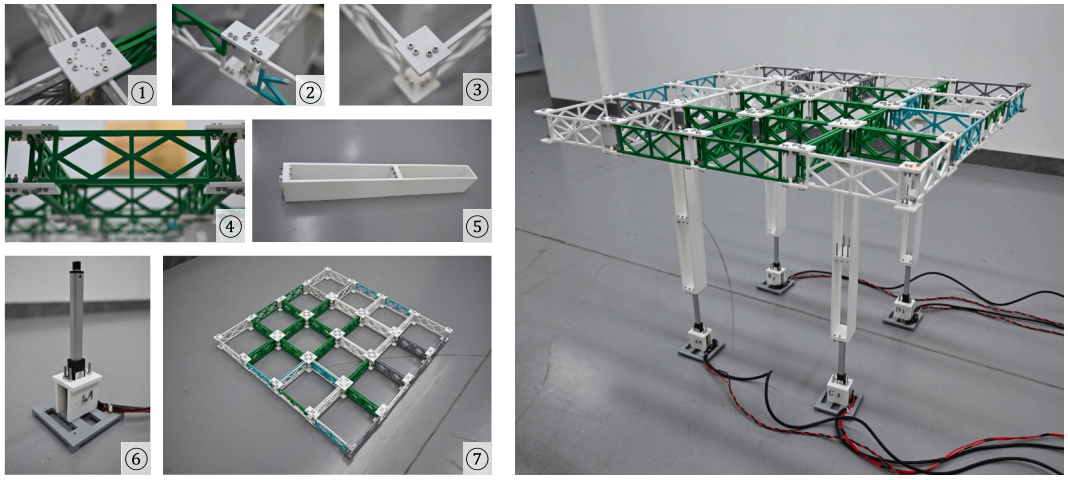


Fig. 3. ABM scale model: (a) Components of ABM scale model, and (b) ABM scale jacking test platform.

seek robust control schemes through the simulation of the ABM model's dynamic system. In each iteration of PPO, the agent interacts with the environment using the current policy to collect data. It assesses the direction of policy improvement by calculating the probability ratio of actions generated by the new and old policies (importance sampling ratio), and strictly limits the magnitude of policy updates by using a clipped objective function to enhance smoothness. The core approach driving this process is to have the agent (representing the optimizer) make numerous attempts in the simulated system environment. Through repeated simulated interactions, PPO explores the PID parameter space and converges to gain values that achieve stable multi-motor synchronization under specified conditions. These optimized parameters provide a high-quality initialization for the subsequent DNN-PID controller, reducing exploration cost and improving convergence efficiency. Unlike direct deployment on physical systems, offline PPO training enables the acquisition of prior knowledge by capturing fundamental control behaviors from simulation.

The following formulas summarize the primary content of the PPO algorithm: (1) is the ratio of the probability of taking action under the current strategy to that under the old strategy in a given state, referred to as the importance weight. (2) is the pruning strategy objective function, which uses the min(·) operation to encourage beneficial updates while suppressing excessive updates that may cause training instability. (3) Value function loss, which minimizes the prediction error of the value network, improves the accuracy of state value estimation, and provides a stable baseline for policy updates. (4) PPO overall objective function, which balances policy loss and prevents local convergence. (5) Generalized advantage estimation, which generates advantage estimates with balanced bias and variance, improves the stability of policy updates.

$$r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta\ old}(a_t|s_t)} \tag{1}$$

$$L_{CLIP}(\theta) = \widehat{E}_t[min(r_t(\theta)\widehat{A}_t\ clip(r_t(\theta), 1-\varepsilon, 1+\varepsilon)\widehat{A}_t)] \tag{2}$$

$$L^{VF}(\phi) = \widehat{E}_t\left[\left(V_\phi(s_t) - V_t^{target}\right)^2\right] \tag{3}$$

$$L^{PPO}(\theta,\phi) = L^{CLIP}(\theta) - c_1 L^{VF}(\phi) + c_2\widehat{E}_t[S(\pi_\theta(.|s_t))] \tag{4}$$

$$\widehat{A}_t^{\ GAE(\gamma,\lambda)} = \sum_{l=0}^{T-t-1}(\gamma\lambda)^l\delta_{t+l} \tag{5}$$

Algorithm 1. PPO algorithm for multi-objective optimization of control parameters

```
Initialize parameters θ, λ, and optimizer
for iteration = 1:N do:
  for episode = 1:M do:
    Collect D trajectories D = [(s,a,r,s')…]
  end for
    Estimate the advantage function Â(s,a)
    for optimization step = 1 to K do:
      for each minibatch of experience (s, a,r,s') do:
        Compute probability ratio:
        r(θ) = πθ(a|s) / πθ old(a|s)
        Define clipped surrogate objective:
        L = min(r(θ) * Â(s,a), clip(r(θ), 1-ε, 1+ε) * Â(s,a)) - λ * R_penalty
      end for
    end for
end for
```

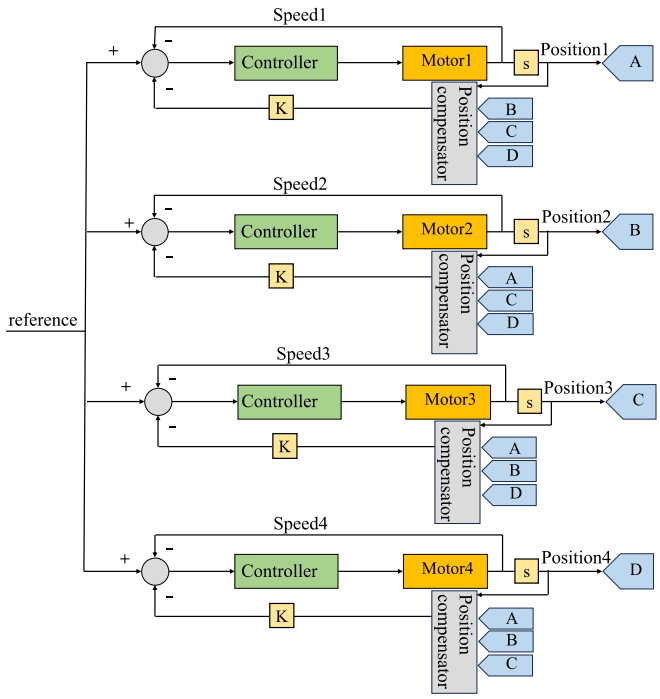


Fig. 4. Advanced control framework of deviation-coupled drive.

3.2.2. *Multi-motor coordination based on deviation coupling*

The jacking power system of ABM is a kind of multi-cylinder synchronous control scenario that is highly sensitive to inter-actuator position consistency, and it also requires speed regulation to satisfy the set jacking velocity requirements. Conventional position-synchronization schemes typically employ a dual-loop PID architecture, wherein an outer position loop generates the control reference for an inner velocity loop. Although effective for stable tracking tasks, this structure focuses solely on position-tracking performance. Under synchronization tasks involving varying loads, the control reference for the velocity loop can exhibit significant fluctuations, which adversely affect the ABM steel platform, the equipment, and materials mounted on it. In this paper, deviation coupling combined with an improved incremental PID scheme is adopted to coordinate the motion of multiple power motors, as illustrated in Fig. 4. In the deviation coupling mechanism, the displacement deviation between each actuator and the group mean is used as the compensation term, which is then fed into the speed regulation loop through a coupling gain. In this process, the deviation compensation quantity is strictly limited to ensure that the speed fluctuates within a certain range from the set value. The adjustment of the compensation gain parameter is to find a balance between the position synchronization control accuracy and the stability of the speed control. This parameter can be adjusted within a certain range according to the needs of equipment use in the actual application process, and the rationality of this adjustment range will be given through the multi-objective optimization of the control effect.

3.2.3. *Transfer learning based on DNN-PID controller*

This paper utilizes MATLAB/Simulink to construct a multi-motor simulation system and trains the neural network to obtain the initial

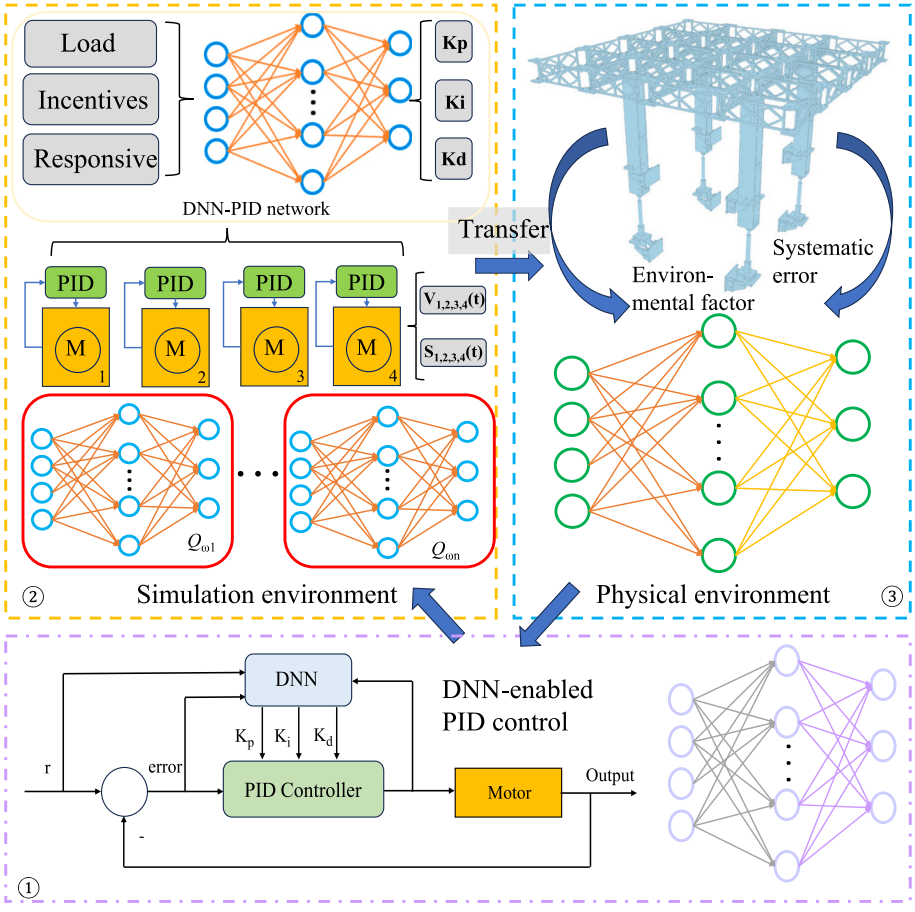


Fig. 5. Transfer learning framework.

Table 1
DNN hyperparameter configurations and definitions.

Item	Value	Description
Number of layers	3	Input-hidden-output
Input features	r_t, u_{t-1}, e_t + bias	Reference target, control output (t-1), error, input bias
Hidden neurons	5	
Output variables	K_p, K_i, K_d	PID gain parameters
W_{ih} size	5×4	
W_{ho} size	3×5	
Hidden activation	tanh	
Output activation	1 + tanh(.)	Range [0,2]
Optimizer	/	Manually defined optimizer
Learning rate	0.01	
Update trigger	et > 15.0	Error threshold for weight update
Gradient clipping	$\Delta \epsilon [-1, 1]$	
Increment clamp	$\Delta u \epsilon [-50, 50]$	Prevent control oscillations
Output limiting	$u \epsilon [45, 100]$	Adapt to the PWM output range

neural network weights. Specifically, a second-order transfer function is employed to approximate the actuator behavior. In practice, even though the scale model in this study uses motors of the same model, there are still differences in physical quantities such as rotational inertia and damping coefficients [44]. Therefore, the motor transfer function in the virtual environment was adjusted based on the physical parameters of the reference motor to account for individual differences, aiming to develop a more adaptive control method. The proposed transfer learning strategy assumes consistency in system structure and control interfaces between simulation and the scaled physical platform, while tolerating discrepancies in parameters, disturbances, and unmodeled dynamics.

realizes the four-motor cooperative jacking driven by a DNN-PID control algorithm in the scale model. Timers, serial ports, and the general-purpose input/output (GPIO) are used to implement specific functions. The primary functions of the proportional model control system, in accordance with its execution logic sequence, are as follows:

(1) Timer reads the encoder pulses of the four actuator motors to obtain the speed and stroke information;
(2) Execute the DNN-PID controller operation module, input the current error, and output the motor speed control signal.
(3) Regulate the PWM duty cycle based on the control signal and connect the driver board to control the motor speed.
(4) Realize cyclic control by timing interrupt, complete the operation of sensor data capture, execute PID operation, output direction, and PWM control signal with a fixed cycle.
(5) The communication module completes the data exchange between the STM32 chip and the host computer, utilizing serial communication to realize the control target setting, real-time feedback of the control effect, and the operation of virtual model driving.

In this process, the elements that have a large impact on the control effect include the gradient tailoring of the DNN and the limitation of the control output. In the initial stage of control, due to the difference in response performance between each power pusher, it is challenging to avoid the scenario of increasing error. The gradient explosion caused by the large error will ultimately lead to the DNN failing. The reasonable gradient trimming used in the initial stage reduces the sensitivity of the network so that it can continuously output the control parameters. In scenarios with large errors, DNN may experience gradient explosion. Additionally, the PI elements of PID controllers may output extremely large control signals under certain control conditions. Therefore, an output limiting strategy is employed to optimize control performance.

Algorithm 2. Pseudo-code for DNN-PID implementation in embedded devices

```
Input: r: reference; actual_error:
Output: output_val: Control signal output
while actual_error>0
    Error <- actual_error and target_val <- r
    Kp，Ki，Kd=DNNwork(target_val，output_val，  Error)
    delta = Kp * (Error - LastError) + Ki * Error + Kd * (Error - 2*LastError + PrevError)
    delta = clamp(delta, -50.0, 50.0)
    output_val = clamp(output_val + delta, 45, 100)
    if |Error| > 15.0
        update_weights of DNN-PID model
    end if
    PrevError <- LastError and LastError <- Error
    return output_val
end while
```

Fig. 5 details the components of transfer learning, where the module ① describes the primary function of the DNN: real-time tuning of PID control parameters, and the module ② details the training of the DNN-PID in a simulation environment. Then, module ③ illustrates the deployment of the control strategy within the physical environment. The network underwent 500 iterative training sessions using identical input-output structures within the virtual environment (trained for 5 s at an update frequency of 100 Hz). Trained under complete constraints, the DNN demonstrated excellent control performance, laying the foundation for its transfer. Relevant details are shown in Table 1.

This paper is based on the STM32 platform to realize the embedded program development of a synchronous jacking control system. It

For the DNN-PID network structure used in this paper, the neural network weights from the input layer to the hidden layer and from the hidden layer to the output layer are first obtained through several simulation trainings. These trained weights are migrated to the embedded environment. During deployment, the weights from the input layer to the hidden layer are fixed, while the weight update logic from the hidden layer to the output layer is increased. Triggering thresholds based on control bias inputs are introduced so that weight updates are activated when large control errors occur, thereby improving the adaptability of the neural network. In the transfer learning process, the trained network parameters obtained in the simulation environment are

migrated to the STM32-based embedded controller deployed on the scale-model platform. The network architecture and input–output interfaces remain consistent between the simulation and physical systems, enabling the controller trained in MATLAB/Simulink to be directly initialized in the embedded environment. During physical experiments, the hidden-to-output layer weights are further updated online according to real-time control errors, allowing the controller to compensate for parameter discrepancies, disturbances, and unmodeled dynamics that exist in the physical system but are not captured in the simulation model.

3.2.4. Training and model evaluation

The construction of the synchronous jacking control system is completed through multiple stages, including model building, simulation, and hardware migration. According to the actual application scenarios of ABM, certain indicators are established to evaluate the performance of the synchronous jacking control. This process particularly focuses on the impact of the control results on the structural safety of ABM. On this basis, some process parameters in the model development and iteration stages are taken into account to achieve efficient optimization iterations. The foundation of the indicator system is the *MSSE* and the *MSE* of ABM jacking. *MSSE* reflects the accuracy and stability when controlling the fixed height of jacking, calculated as shown in Eq. (6), determined by the maximum error between multiple motors and the control target; MSE reflects the synchronization performance of multiple motors, directly affecting the structural safety of ABM, calculated as shown in Eq. (7), by the range of the control results of multiple motors. All the above indicators are instantaneous values at time points. The *ASE* is calculated to reflect the control effect of the jacking process, as shown in Eq. (8), by averaging all the MSE values at the sampling.

$$MSSE = \max[|\mathscr{G} - r_i(t)|], i = 1, 2, \cdots, n \tag{6}$$

$$MSE = \max r_i(t) - \min r_i(t), i = 1, 2, \cdots, n \tag{7}$$

$$ASE = \frac{1}{N}\sum_{i=1}^{N} MSE(i) \tag{8}$$

where $\mathscr{G}$ is the control target of the route, $r_i(t)$ is the final route of each motor, n is the count of motors, and N is the count of steps of time.

The architecture of the PID control model is based on the Proximal Policy Optimization (PPO) reinforcement learning algorithm implemented in Python, which is specifically designed for multi-objective optimization and is used to optimize PID parameters in a multi-dimensional continuous space. The PPO agent interacts with the simulated jacking environment to evaluate the utility of different control strategies multiple times and learn the best control strategy. The reinforcement learning method's reward function is set up to be more adaptable to multi-objective optimization problems than traditional methods.

The reward function is designed as the weighted sum of multiple indicators, integrating three key performance indicators crucial for synchronous hydraulic jacking: cumulative synchronization error, cumulative tracking error, and cumulative control energy consumption (reflecting the stability of the control process). The calculation items of the reward function are shown in Eqs. (9), (10), and (11). Reward weights are used to balance these competing objectives, and the factors influencing the weight setting include the importance of the indicators and the order of magnitude of the calculation. The reward function drives the PPO model to optimize the control strategy with the core focus on synchronization error control. Reward is used to guide the modification and improvement of reinforcement learning methods. Meanwhile, its convergence trend and speed can be utilized to evaluate the advantages and disadvantages of different reinforcement learning methods. The convergence ability of reinforcement learning methods in different scenarios is influenced by many factors.

$$f_1(x) = \frac{1}{T}\int_0^T t|MSSE(t)|dt \tag{9}$$

$$f_2(x) = \frac{1}{T}\int_0^T t|MSE(t)|dt \tag{10}$$

$$f_3(x) = \sum_{i=1}^{4} u_i^2 \tag{11}$$

where u_i^2 is the energy cost of each control process.

Based on the control strategy obtained from the multi-objective optimization of the PPO algorithm, the DNN-PID compensation control network was developed. As an adaptive closed-loop controller, the training and verification of the DNN-PID network were carried out in MATLAB simulation. This DNN is used with multiple sampled reference values, deviations, and control outputs as network inputs, and the output sets the PID parameters. The backpropagation of errors in the DNN network is replaced by the backpropagation of control deviations, that is, the input-output coupling of the network is improved for this task. At the same time, the synchronization error simulating the jacking process is used to evaluate the effectiveness of the network.

$$\tau = \frac{v_{nominal} - v_{actual}}{t_{response}} \tag{12}$$

where $v_{nominal}$ is the motor speed under normal conditions, v_{actual} is the motor speed under load, and $t_{response}$ is the response time constant.

The priority is to monitor the actual synchronization error curve of the entire jacking process during operation. This encompasses analyzing transient responses under step commands and performance tracking within coordinated trajectories. It further includes adaptability to non-uniform loads or simulated component failures, along with overall stability and robustness demonstrated across the entire operating range. Consequently, the core evaluation metric centers on the accuracy of synchronization errors within hardware systems deploying the DNN-PID approach.

3.3. Digital twin implementation and interface

To facilitate the necessary data exchange and provide the required interfaces for future extensions of this research, this paper proposes a digital twin system architecture designed for ABM synchronous jacking experiments as illustrated in Fig. 6. The architecture primarily consists of a physical entity layer, a data communication layer, a digital model layer, and an application service layer. As shown in the figure, the physical entity layer comprises a scaled ABM experimental platform and its sensor system, which are used to collect the dynamic state of the physical model. The data communication layer handles the upload of sensor data and the transmission of control commands, through the embedded communication interface described in Section 3.1, which is implemented based on serial communication between the STM32 controller and the host computer. The digital model layer performs mapping and visualization, while the application service layer provides functions such as experiment monitoring, logging, and control command management, thereby forming a digital twin system framework that supports experimental operation management and data analysis. The ABM virtual model is used to display the real-time status of the jacking process and support safety risk analysis and control. Its establishment is based on the structure of the scale model entity and the analysis of the jacking motion state. The virtual model is completed by SolidWorks to model and assemble complex components, and it also serves as a blueprint for the fabrication of the scale model components. The difference between the virtual model and the three-dimensional design drawing lies in the integration of components without relative

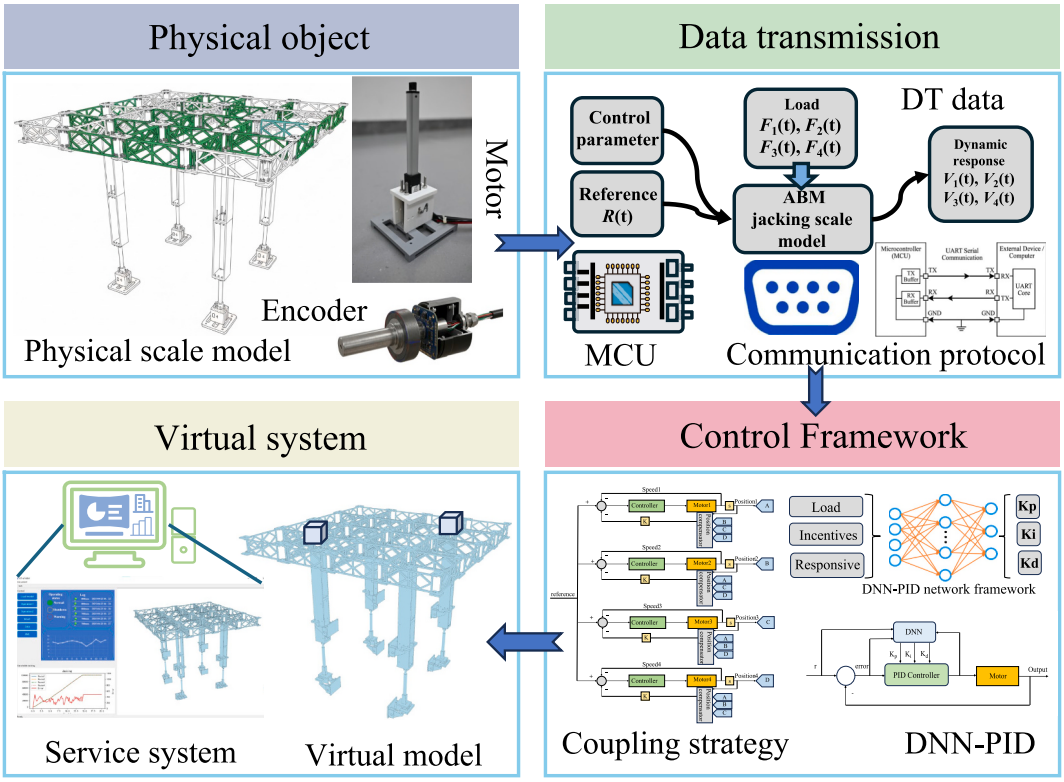


Fig. 6. Digital-twin feedback loop process.

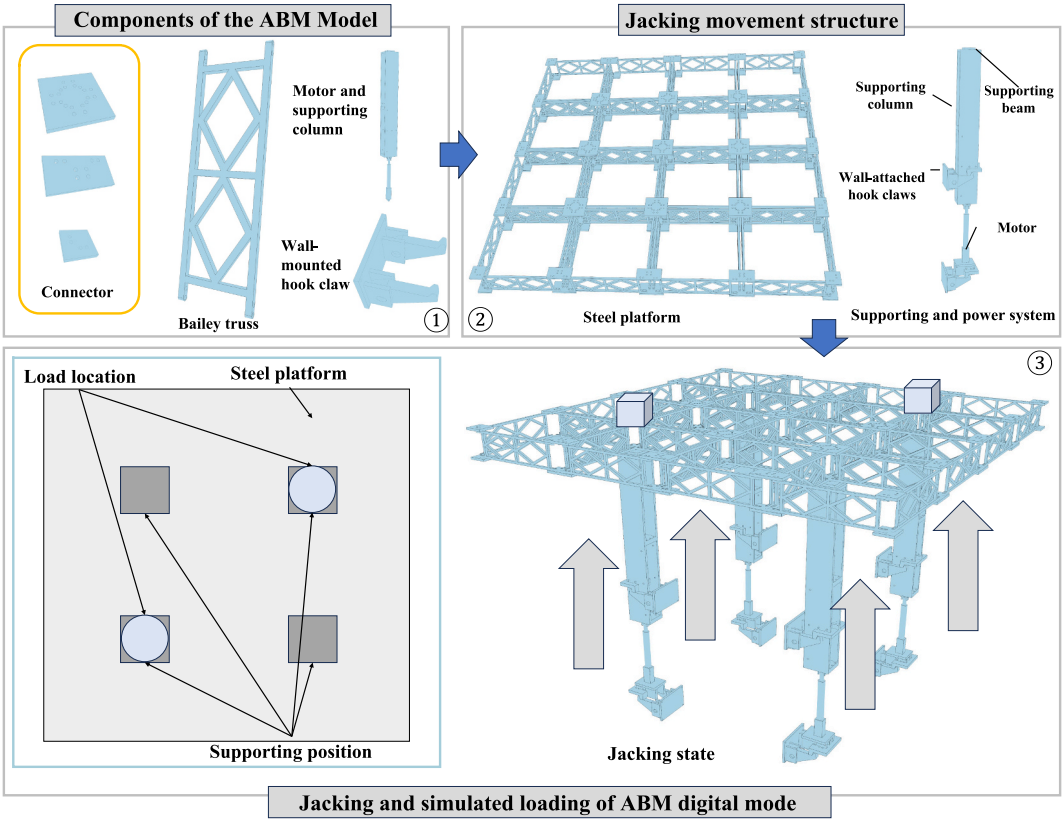


Fig. 7. Components of the virtual model, motion module, and loading and jacking illustration.

displacement during the jacking process. Components such as the Bailey truss, connectors, supporting columns, motors, and extension sections are reflected in the virtual model by two major structures: the steel platform and the supporting and power system.

The utility demonstration of the data-driven digital twin system is shown in Fig. 7. During the jacking process of the scale model of ABM, the main motion mode is the extension process of four motors used as the power system. Each set of digital twin data includes the speed, stroke, and timestamp of the four motors, enabling synchronous visualization and real-time monitoring of the operating posture. In addition, the digital twin issues target displacement commands and start–stop instructions to the scale model, forming a supervisory closed loop that coordinates experiment execution while maintaining data consistency between virtual and physical domains. Furthermore, during the future deployment of this control strategy onto real ABM systems, the DT will be more comprehensively integrated into the control loop, encompassing both real-time monitoring and online learning capabilities.

4. Experimental study

To demonstrate the effectiveness of the control strategy employed in this study, empirical analysis is conducted using relevant data from an ABM project in China. The high-rise building project is located in Wuhan, China. The building has 119 floors above the ground with a height of 380 m. More details of the data, implementation details, and the analysis of the test results are illustrated in Sections 4.1, 4.2, and 4.3, respectively.

4.1. Data and resources

The real-life case corresponding to the scale model of the ABM in this paper is located in Wuhan, China. The project consists of an 80-storey tower above the ground and a 4-storey tower below the ground. The height of the concrete structure of the tower is 380 m, and the standard floor height is 4.4/4.5 m. The construction adopts the fourth-generation aerial building machine - lightweight integrated platform system. It optimizes the construction process of this building through the key technological advantages, including standardized design, industrialized production, integrated assembly construction, and intelligent equipment integration. As a result, the system enables the cyclic construction of one standard floor every four days and the climbing of one structural floor in 60 min. The construction process is optimized through standardized design, industrial production environment, integrated assembly construction technology, intelligent equipment integration, and other key technical advantages, realizing the cyclic construction of one standard floor in four days and the overall climbing of one structural floor in 60 min, which reduces the number of construction personnel by 50% and improves the working efficiency by 30% compared with that of the mold-climbing platform. The upper platform of this ABM has an area of about 800 m^2, equipped with two tower cranes and two fabrication machines. It includes three operating layers: the reinforcement binding layer, the concrete pouring layer, and the concrete curing layer, enabling “assembly line” construction operation. The top layout and jacking control system of ABM are shown in Fig. 8.

This ABM entity has 12 cylinders as its jacking power system, and the steel platform is supported by 12 sets of support systems containing these cylinders. To study the synchronous jacking control of this type of ABM system, develop adaptable cooperative control algorithms, and be able to verify the validity and superiority of the algorithms practically, this paper simplifies the number of power and support systems to 4 sets employing scale-down modeling and considering the research needs and actual conditions, and designs and establishes a scale model of a 1:25 ABM as shown in Fig. 9, and deploys an intelligent algorithm for synchronous jacking control on the model, which is used as the power



(a) (b) (c)

Fig. 8. ABM for the Wuhan Yangtze River Center: (a) Overall view of ABM, (b) ABM construction platform and its unevenly distributed loads above, and (c) Jacking control system.

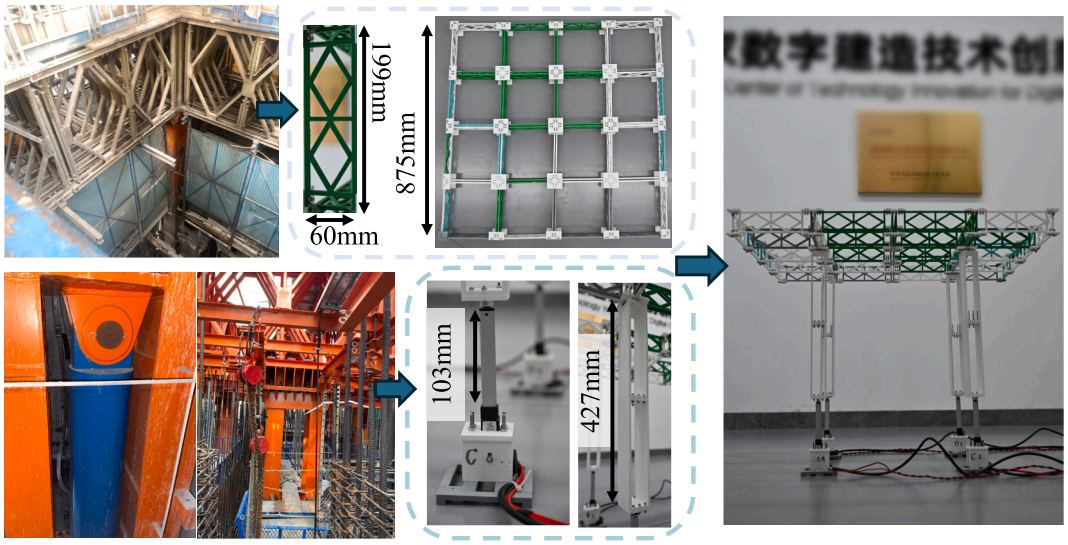


Fig. 9. Using the principle of similarity to construct a geometric similarity scale model for the jacking test platform based on the ABM employed in the actual construction project.

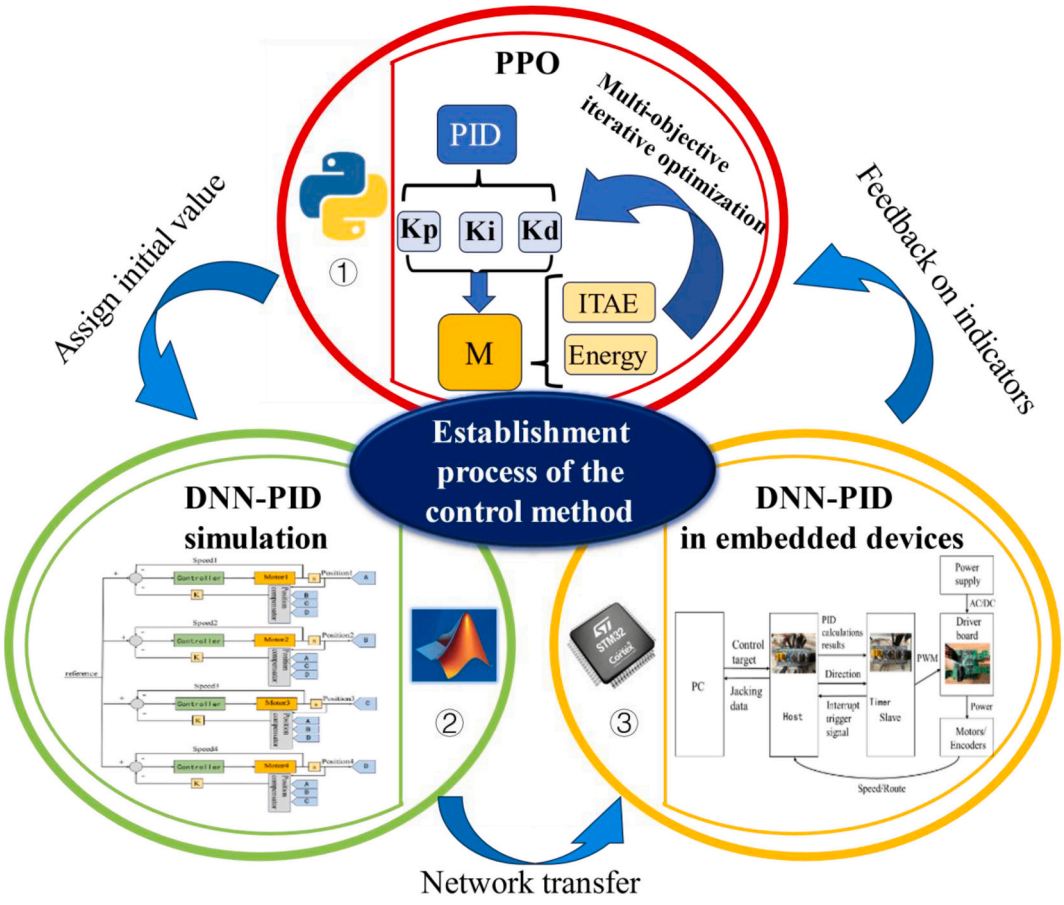


Fig. 10. Establishment and deployment of control method.

system of this aerial building machine. Deploy the intelligent algorithm of synchronized jacking control, and execute the jacking test with the scale model.

4.2. Implementation details

In this paper, the ABM synchronized jacking control system was built in three stages, as shown in Fig. 10: equation simulation, electrical simulation, and physical deployment for multi-objective driven optimization of the PID control parameters, iterative updating of the weights of the DNN network, and migration and deployment of the control network, respectively. Different objective functions are used to guide the optimization of the parameters in the three phases. The parameter transfer for establishing a closed-loop control system can be described as

Table 2
PPO hyperparameter configurations.

PPO Hyperparameter	Symbol	Value
Discount factor	γ	0.99
GAE parameter	λ	0.95
Learning rate	α	3.0×10^{-4}
Clipping parameter	ε_{clip}	0.2
Epochs per update	/	10
Batch size	/	128
Rollout steps	/	2048
Entropy coefficient	c_{ent}	0.01
Value loss coefficient	c_v	0.5
Max gradient norm	/	0.5
Exploration mechanism	/	Stochastic policy with entropy regularization (ent_coef = 0.01)
Policy network architecture	/	2 hidden layers (64, 64)
Value network architecture	/	2 hidden layers (64, 64)
Total training timesteps	/	120,000

Table 3
Environmental configurations for PPO optimization.

Item	Symbol	Value/ Description
Number of actuators	N	4
Control cycle	dt	0.01 s
Episode duration	T	5.0 s
Action variables	/	PID gains [Kp, Ki, Kd]
Action bounds	/	Kp∈[0.01, 0.5], Ki∈[0.001, 0.1], Kd∈[0.001, 0.1]
Observation dimension	/	20
Synchronization	K_{sync}	0.01
Reward weights	/	$\omega_v = 0.3$, $\omega_{sync} = 0.7$, $\omega_u = 0.001$

follows. The optimal PID parameters obtained from the PPO are used to establish the DNN-PID. Subsequently, the optimal network parameters obtained through multiple iterations of the DNN are transferred to the physical system. Then, the operational feedback from the physical system is used to optimize the control strategy, forming a closed-loop system.

The PPO algorithm is implemented in python platform and the multi-motor system is simulated using Euler equations; Simulink is used to build the multi-motor coupled system containing DNN-PID control method and the function of the DNN network is realized based on its S-function; in the physical system, the operation of the DNN-PID network, including its numerical computation and the required data and signal generation, is based on the embedded devices, encoders, driver boards, and other hardware implementation. The establishment process of the ABM synchronous jacking control system includes: the PPO algorithm performs multiple optimization searches to obtain the PID control parameter used for the multi-motor system. The hyperparameter configurations for the PPO algorithm are summarized in Table 2. Table 3 presents the environmental setups used for PPO optimization. The parameters obtained from PPO are used for the initial value of the control parameter of the DNN-PID simulation network. It is updated by multiple iterations of the weights of the DNN, and then deployed to the physical system through the transfer learning method, so that the control system is deployed, and the adaptive updating of the network can be continued during the control process. Table 1 presents the architecture and hyperparameters of the DNN. Specifically, parameters such as output limits, update thresholds, and increment clamps are designed to enhance system stability. These parameters are determined through simple grid searches or based on the practical constraints of the physical environment. The PPO with Simulink model is deployed on a standard PC with 11th Gen Intel(R) Core (TM) i7–11,700 CPUs, NVIDIA GeForce RTX 3050 GPU cards, and 16 GB RAM for training and simulation. Fig. 11 shows the training process and test results of the PPO algorithm. In the simulation environment, the DNN was trained for 5 s at an update frequency of 100 Hz, resulting in a total of 500 iterations.

The deployment of the control method in the scale model is based on a complex hardware architecture. As shown in Table 4, the most central master control device is the STM32F103C8T6. Hardware resources such as timers, serial ports, GPIO ports, etc., are used for the implementation of the control functions, and the standard library functions provided by STM are implemented to manipulate the registers in order to control the corresponding hardware. The implementation of parallel control

Table 4
Hardware specifications of scaled ABM experimental platform.

Component	Type	Key specification	Function
Microcontroller	STM32F103C8T6	72 MHz, 64 KB Flash	Main control unit
Motor	Mini electric actuator Brushed DC motor	Stroke: 100 mm Speed: 4 mm/s Max thrust: 120 N	Drives jacking motion
Encoder		Resolution: 7 PPR	Displacement feedback
Control cycle	Loop interruption	Period: 300 ms	Controller sample

Note: the maximum thrust of this motor model varies with speed: at 4 mm/s, the maximum thrust is 120 N; at 6 mm/s, it is 80 N; and at 15 mm/s, it is 40 N.

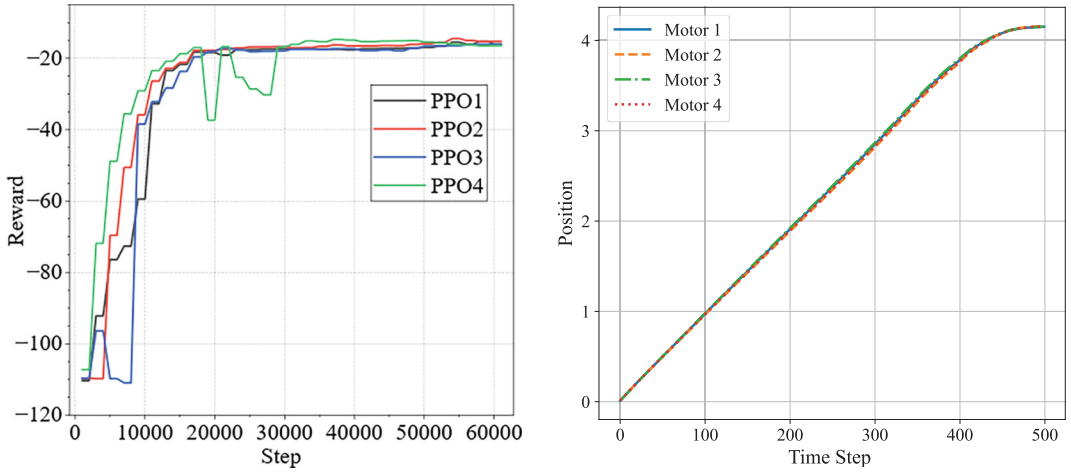


Fig. 11. PPO training process and test results: (a) Reward and (b) Test results.

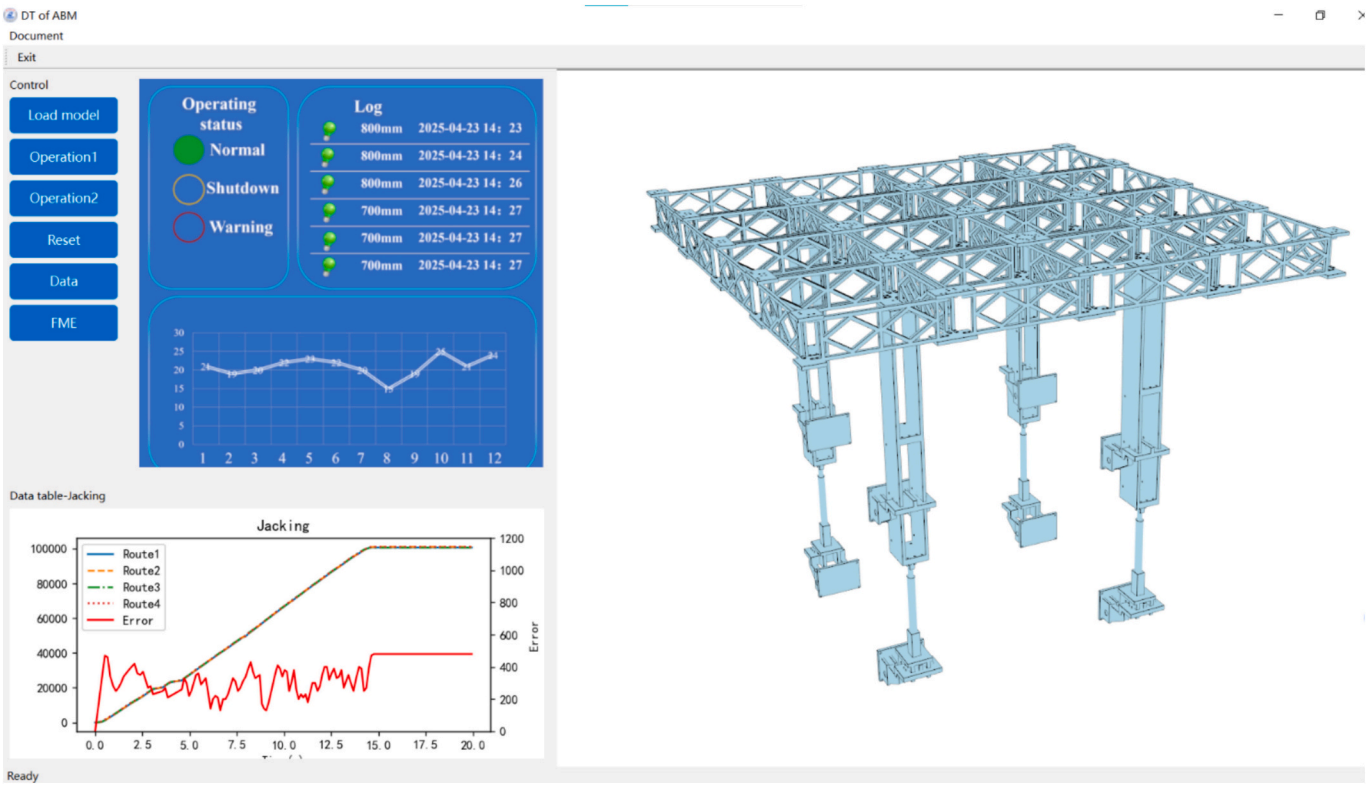


Fig. 12. Digital twin platform for ABM control service interface.

functions, including DNN, incremental PID calculation, data acquisition processing, and packetization for sending, is written in the C language. A master-slave architecture is designed to solve the bottleneck of timer resources. The master carries out data acquisition, processing, and communication with the host computer and DNN-PID calculation; the slave carries out PWM signal generation and cyclic control, and serial communication is used to transfer the control data between the master and the slave. In the physical system, the execution period of the control loop is 300 ms. The DNN initially obtained through simulation was adaptively and iteratively updated during the jacking tests of the physical system for a total of 52 epochs. Specifically, the encoder resolution is seven lines, meaning that the motor outputs seven pulses per revolution. Within the context of this study, the displacement mapping relationship is established as 0.00074 mm per pulse.

For each operating condition and control strategy, five independent jacking experiments were conducted to ensure repeatability and statistical reliability. In each trial, the termination condition of the jacking process was defined by a preset target stroke. As the measured stroke approached the target value, a deceleration logic was automatically triggered to ensure a smooth and controlled stop of the actuators. The experimental procedure followed a consistent sequence: (a) the loading coefficient was configured at the firmware level of the embedded control system; (b) the target stroke was specified and the start command was issued via the DT system; (c) during the jacking process, the embedded controller executed the synchronization control algorithm while motion and synchronization data were continuously collected and logged in real time; and (d) upon completion of the jacking operation, the recorded data were exported for subsequent statistical analysis and performance

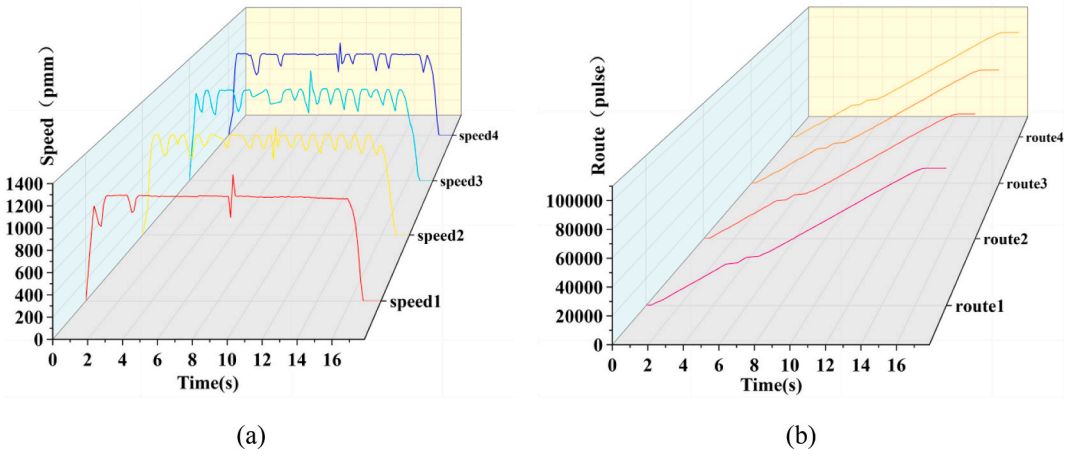


Fig. 13. Jacking data presentation: (a) Speed and (b) Route.

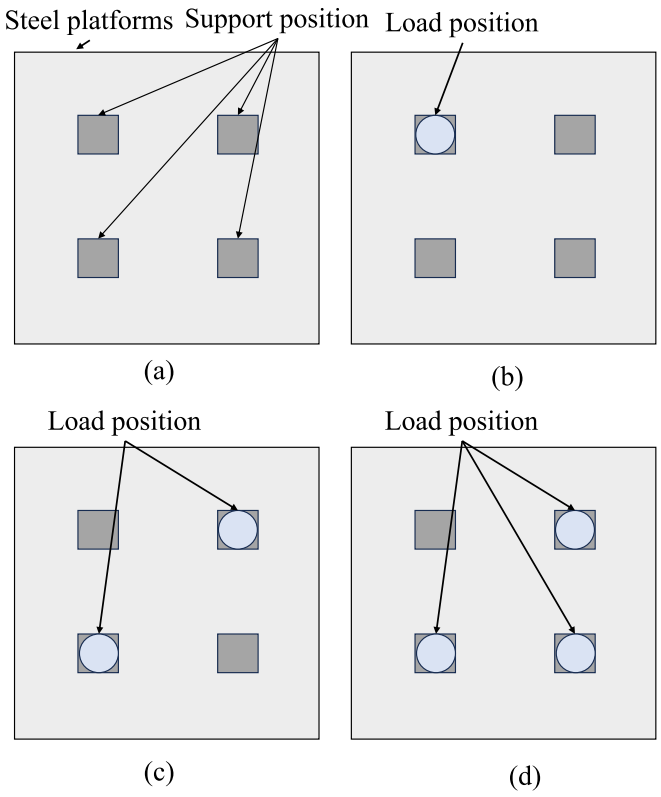


Fig. 14. Loading design for jacking tests: (a) Support schematic, (b) Single point loading, (c) Diagonally symmetrical load, and (d) Three-point loading.

evaluation. The control loop performs data acquisition and processing every 300 ms, which introduces an inherent delay in the twin data. Additionally, serial communication and data processing by the host computer contribute to system latency. According to calculations, the twin delay is 1–1.5 s.

This study constructed a digital twin system based on a proportional model of the physical entity, enabling data exchange between the host computer and the main control board via serial communication. The system receives real-time motor pulse and jacking height data, which is parsed by the backend to drive the virtual model, store operational status, and monitor the physical entity. The service interface, as shown in Fig. 12, developed using PyQt, provides data visualization, operational log recording, and safety alert functions. The Unity3D platform is employed for virtual mapping and simulation feedback. Centered on twin data, this system supports control strategy analysis and optimization through virtual-physical interaction. The jacking synchronous control problem of ABM is concerned with the stroke synchronization between each power mechanism, which is directly related to the level of the steel platform and further affects the equipment safety. The form of jacking data obtained during the control process is shown in Fig. 13. These data include timestamps and the travel and speed information of the four jacking motors, which is a class of time series data with causal correlation. These data are both inputs to the intelligent control algorithm for the iterative optimization of the network and outputs of the control effect for calculating and analyzing the effect of the control algorithm.

In this paper, jacking tests demonstrate the superiority of intelligent synchronization control algorithms compared to traditional control methods, select the best multi-motor cooperative control algorithm to build the control strategy network based on the test results, and verify the synchronization performance of the multi-motor cooperative jacking system on the ABM scale model under various working conditions. For the non-uniform loads faced by the ABM under complex working conditions, this study uses the principle of equivalent load on the scale model, i.e., limiting the maximum speed of the specified motors $v_{max} = k \cdot v_{nominal}$ ($k \in (0,1]$) by the control firmware to establish the mapping relationship between the speed constraints and the loading torque, which is theoretically supported by Eq. (12). The loading scheme is shown in Fig. 14, where single-point loading, diagonal loading and three-point loading are performed on the scale model to simulate complex working conditions.

4.3. Analysis of results

In this part, the jacking test based on the scale model platform is carried out to verify the effect of the intelligent synchronous control algorithm used for ABM jacking control. Firstly, the jacking test without adding intelligent algorithm is carried out, and the results are shown in

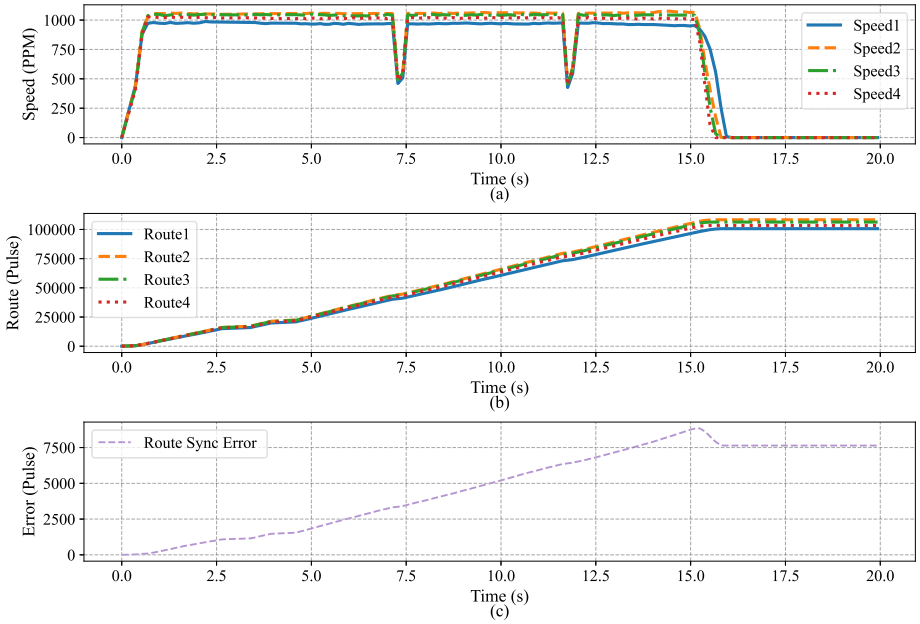


Fig. 15. Jacking tests without intelligent algorithms: (a) Multi-motor speed, (b) Multi-motor route, and (c) Synchronization error.

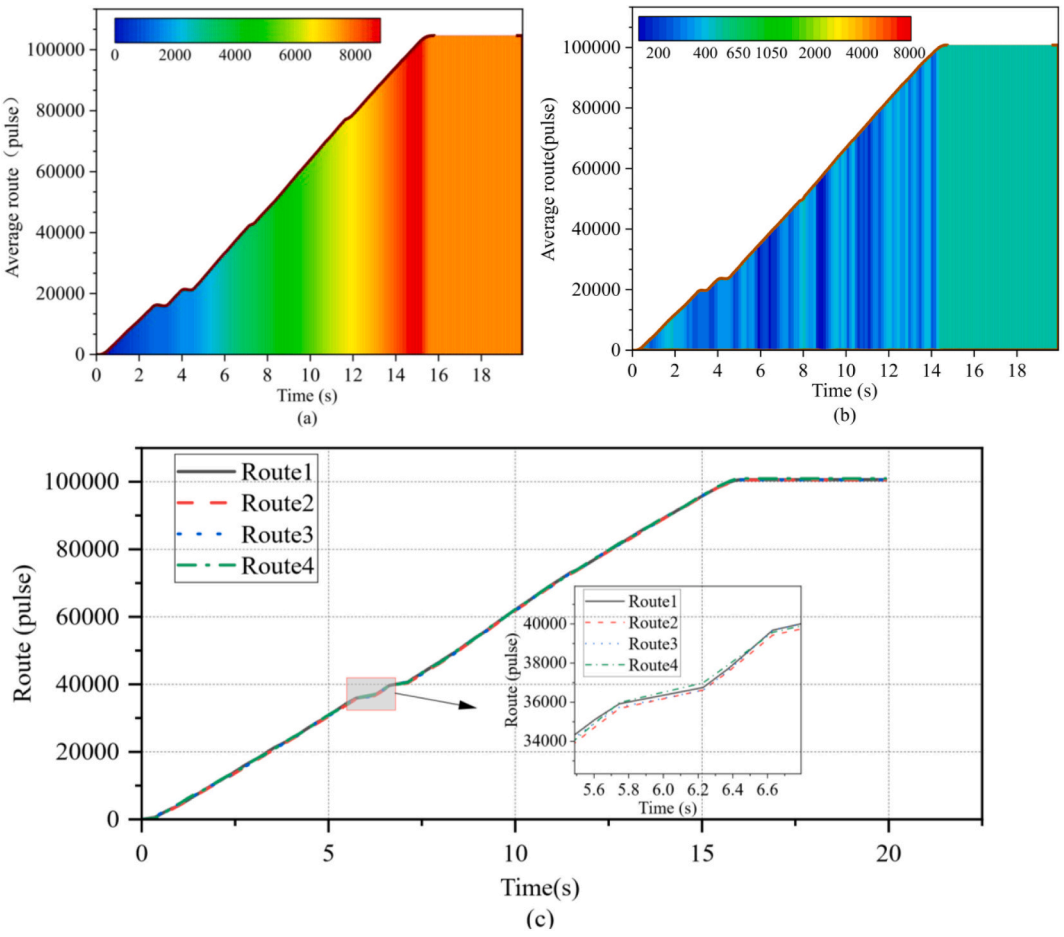


Fig. 16. Effect of adding an intelligent synchronization algorithm on jacking synchronization error: (a) Synchronization error without intelligent algorithms, (b) Synchronization error with intelligent algorithms, and (c) Jacking process with intelligent algorithms.

Fig. 15. Then the jacking test with adding intelligent algorithm is carried out, and the results are shown in Fig. 16. Then the simulated loading test under various working conditions is carried out to verify the robustness of the control algorithm in the scaled experimental environment, and the results are shown in Figs. 17–18. Finally, the jacking test driven by several different coupling methods is carried out to verify the effect of the modification, and the results are shown in Figs. 19–20. The results are analyzed as follows.

(1) Without the intelligent control method, the synchronization error exceeded the limit value by 13 times, mainly due to the response differences of the physical system. An uncoupled jacking test driven by equivalent control and conventional PID methods was conducted to investigate the response characteristics of the scale model power system. As shown in Fig. 15, the four motors exhibit significantly different responses under the same excitation conditions. Motor #2 reaches the highest speed, while motor #1 operates at the lowest speed and is significantly slower than the other motors. The difference in motor responses leads to the accumulation of velocity errors and the dispersion of travel synchronization errors during the jacking process. During the test, the synchronization error increases continuously as the motors move toward the target stroke. When the first motor reaches the preset stroke, the synchronization error reaches its maximum value. Subsequently, the remaining motors continue moving toward the target position. As shown by the speed curves, motor #1 has the longest running time, and its delayed completion partially compensates for the accumulated error, causing the synchronization error to decrease at the end of the process. For a jacking stroke of 800 mm, the allowable synchronization error is limited to 5 mm. In the scaled experiment, the target stroke of 100,000 pulses corresponds to a maximum allowable synchronization error of 625 pulses (0.463 mm). However, the MSE during the test reaches 8200 pulses (6.07 mm), far exceeding the safety threshold. Such excessive synchronization errors may lead to structural deformation of the steel platform and instability of the support system. Therefore, it is necessary to develop improved control strategies to enhance both the tracking performance of individual motors and the coordination capability among multiple actuators.

(2) The average MSE after using the intelligent algorithm is 0.377 mm (under 5 tests). The average error reduction is 5.87 mm, with a 95% confidence interval of [5.42, 6.32]. The method can significantly reduce the synchronization error during jacking thereby improving the operational stability of the system. DNN dynamically tunes the PID control parameters during the jacking process. Deviation coupling uses the average of the four-motor stroke as the stroke control target for individual motors and calculates the stroke synchronization deviation to compensate for the DNN-PID control target. Fig. 16 shows the jacking test process before and after adding the intelligent synchronization control algorithm, where Fig. 16(a) is the jacking synchronization error without adding the intelligent synchronization control algorithm, and Fig. 16(b) is the jacking synchronization error after adding the intelligent synchronization control algorithm. The MSE during the jacking process was measured as 0.407 mm, which remains within an acceptable range,

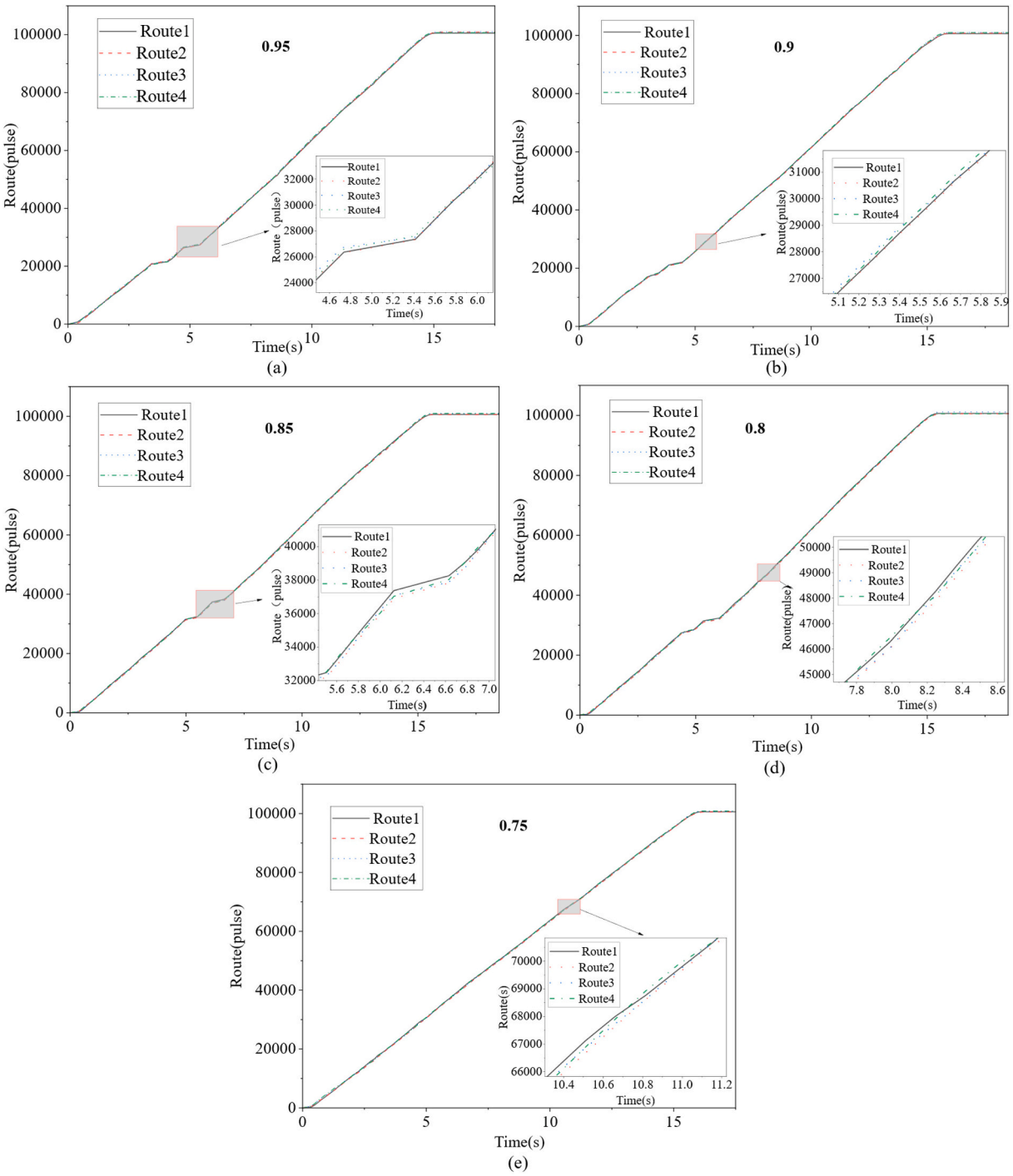


Fig. 17. Route curve in loaded jacking tests: (a) Loading factor: 0.95, (b) Loading factor: 0.9, (c) Loading factor: 0.85, (d) Loading factor: 0.8, and (e) Loading factor: 0.75.

thereby preliminarily verifying the effectiveness of the proposed method. In a single jacking test, the maximum synchronization error generally appears in the system to obtain excitation after the initial sampling of the sensor, the error is caused by the initial response characteristics of the system, after the intelligent algorithm to correct, the error in the fluctuations in the decline and amplitude decreases and gradually tends to be stable. Fig. 16(c) shows the trend of the multi-motor travel during a single jacking process. After receiving the trigger command and the control target, the four motors run synchronously under the control of the intelligent algorithm, as shown in the zoomed-in diagram. Lagging or advancement of individual motors in a multi-motor system is corrected promptly. MSSE is 0.703 mm (950 pulses), in the range of 5%, to achieve the control objectives.

(3) The average MSE under multiple loading conditions is 0.417 mm, with a maximum value of 0.451 mm, all falling within the specified tolerance range. Fig. 17 details the jacking process for single-point loading with different loading coefficients. As shown in Table 5, the mean value of MSEs of 0.376 mm, 0.431 mm, 0.432 mm, 0.406 mm, and 0.436 mm for single-point loading with loading factors of 0.95, 0.9, 0.85, 0.8, and 0.75, and the target stroke setting of 74 mm

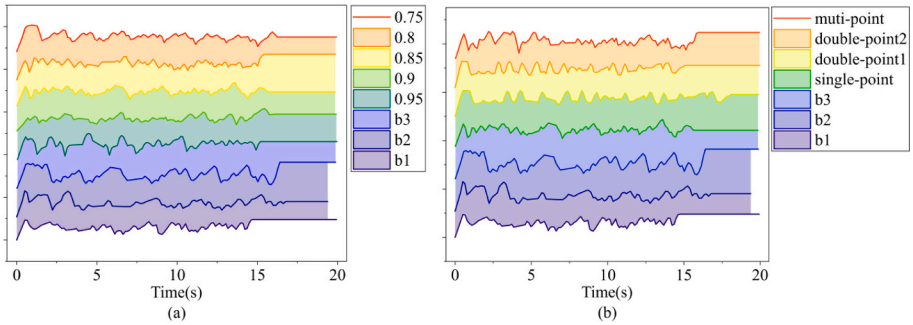


Fig. 18. Characterization of synchronization error fluctuations in loaded jacking tests: (a) Empty and single-point loading and (b) Empty and multipoint loading.

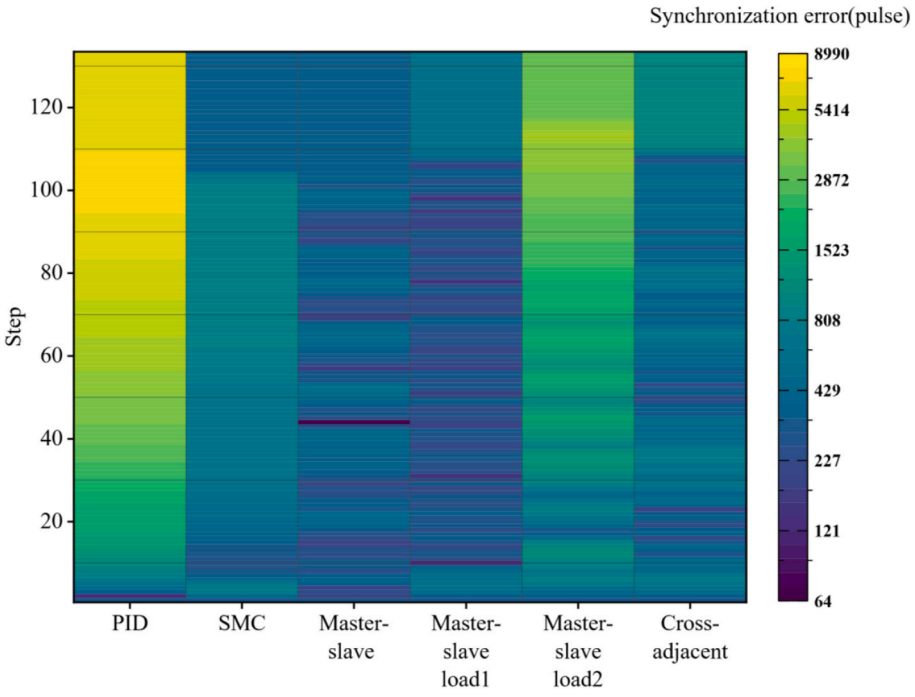


Fig. 19. Comparison of control effects using multiple different control methods in the ABM scale model, shown with synchronization error.

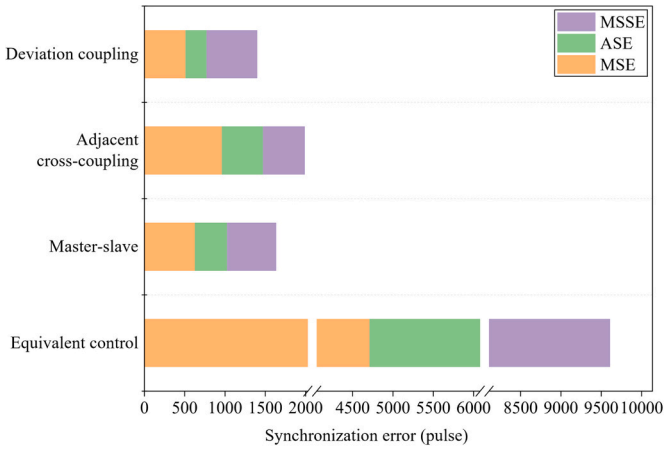


Fig. 20. Comparison of indicators of different control methods.

(100,000 pulses), respectively, are all within the limits. Fig. 18(a) shows the synchronization error fluctuation of jacking under no load and different single-point loading coefficients; Fig. 18(b) shows the synchronization error of jacking under no load and multi-point loading. In this paper, the data with the largest synchronization error are selected from multiple tests under the same loading condition to reliably describe the control effect. The fluctuation trend of synchronization error under different loading factors and conditions shows consistency, i.e., larger synchronization error occurs in the start and stop phases of the jacking. The MSEs of single-point loading, diagonal loading, adjacent loading, and three-point loading are 0.434 mm, 0.428 mm, 0.408 mm, and 0.441 mm, respectively, all within the specified range. The synchronization error in the start phase is caused by the response difference between the motors, and it can be corrected in a relatively short period. The specific data metrics are presented in Table 5. These results indicate that the proposed control strategy maintains synchronization performance within the specified limits under different loading conditions on the scale model without human intervention, demonstrating its adaptability to complex and non-stationary jacking scenarios.

(4) The improved deviation coupling algorithm achieved an average MSE of 0.377 mm, significantly lower than the 3.485 mm, 0.463 mm, and 0.709 mm errors recorded for equivalent control, master-slave control, and adjacent cross-coupling control. A comparison

Table 5
Indicator statistics under multiple loading conditions.

Indicator	Loading	Mean	Max	Min	Std.	mean ± SD ($n = 5$)
MSE	S1	0.377	0.407	0.34	0.028	0.377 ± 0.028
	S2	0.376	0.422	0.333	0.043	0.376 ± 0.043
	S3	0.431	0.437	0.422	0.006	0.431 ± 0.006
	S4	0.432	0.444	0.414	0.013	0.432 ± 0.013
	S5	0.406	0.422	0.392	0.015	0.406 ± 0.015
	S6	0.436	0.451	0.414	0.015	0.436 ± 0.015
	S7	0.434	0.451	0.422	0.011	0.434 ± 0.011
	S8	0.428	0.444	0.414	0.014	0.428 ± 0.014
	S9	0.408	0.414	0.392	0.010	0.408 ± 0.01
	S10	0.441	0.451	0.429	0.008	0.441 ± 0.008
ASE	S1	0.193	0.215	0.164	0.023	0.193 ± 0.023
	S2	0.197	0.216	0.180	0.013	0.197 ± 0.013
	S3	0.291	0.321	0.257	0.025	0.291 ± 0.025
	S4	0.319	0.339	0.300	0.015	0.319 ± 0.015
	S5	0.287	0.318	0.252	0.025	0.287 ± 0.025
	S6	0.270	0.297	0.258	0.016	0.270 ± 0.016
	S7	0.291	0.305	0.269	0.016	0.291 ± 0.016
	S8	0.29	0.321	0.252	0.027	0.290 ± 0.027
	S9	0.283	0.301	0.263	0.015	0.283 ± 0.015
	S10	0.285	0.298	0.271	0.012	0.285 ± 0.012
MSSE	S1	0.468	0.533	0.377	0.08	0.468 ± 0.08
	S2	0.428	0.474	0.363	0.041	0.428 ± 0.041
	S3	0.357	0.518	0.252	0.106	0.357 ± 0.106
	S4	0.311	0.407	0.229	0.075	0.311 ± 0.075
	S5	0.289	0.385	0.229	0.067	0.289 ± 0.067
	S6	0.247	0.348	0.185	0.061	0.247 ± 0.061
	S7	0.266	0.340	0.178	0.058	0.266 ± 0.058
	S8	0.257	0.333	0.133	0.087	0.257 ± 0.087
	S9	0.257	0.303	0.170	0.054	0.257 ± 0.054
	S10	0.256	0.326	0.185	0.054	0.256 ± 0.054

of metrics across control strategies is shown in Table 6. A more advanced sliding mode control (SMC) method was also used for comparison. However, its parameter tuning and performance are limited by the accuracy of the model, and the detailed metrics are shown in Table 7. Even after manual parameter tuning of SMC in a physical environment, the proposed DNN-PID method still achieved a lower average MSE, outperforming SMC by 0.28 mm, with a 95% confidence interval of [0.251, 0.309]. Fig. 19 presents the entire process of synchronization error for different methods controlling the jacking in a color mapping form. The synchronization error of the jacking process driven by equivalent control and traditional PID is significantly greater than that of other methods. A logarithmic mapping is used to map the synchronization error values to the color list to reflect more details of the small error range in the figure. Among the other methods, the adjacent cross-coupling control method does not show a clear advantage, and its synchronization error level is higher than that of master-slave control and deviation coupling control. As shown in Fig. 20, master-slave control shows performance comparable to that of deviation coupling control under certain conditions, which is worth noting. For this method, the synchronization performance can still be maintained at a high level when the main motor is loaded. However, when the slave motor is loaded, the MSE shows a certain divergent trend and significantly exceeds the threshold. This comparison shows that the external load form has a significant impact on the synchronization performance of the multi-motor system controlled by the master-slave control scheme. The tracking errors under different control strategies remain similar and show no significant differences.

5. Discussion

There are some more critical steps in the design, implementation, and deployment of intelligent control methods that are worth exploring. Transfer learning exists in intelligent synchronization control methods in order to connect the simulation and physical model space, and is expected to be used to compensate for the differences in motor response characteristics between the simulation and physical environments. This paper compares the control effect before and after adding transfer learning, and the results are shown in Fig. 21. Then, this paper compares the effect of PPO, DDPG, and SAC reinforcement learning algorithms in multi-objective optimization of PID parameters with the training efficiency, and the results are shown in Fig. 22. All the results are discussed in detail as follows.

(1) An ablation study shows that incorporating transfer learning reduces the average MSE by 29%. The synchronization error of multiple jacks incorporating transfer learning was 0.414 mm,

Table 7
Comparison of metrics for benchmark control algorithms.

Indicator	Method	Mean	Max	Min	Std.	mean ± SD (n = 5)	P-value (n = 5)
MSE	SMC	0.657	0.688	0.599	0.037	0.657 ± 0.037	5.51E-06**
	↓ DNN-PID	0.377	0.407	0.34	0.028	0.377 ± 0.028	
ASE	SMC	0.45	0.474	0.406	0.027	0.45 ± 0.027	3.04E-06**
	↓ DNN-PID	0.193	0.215	0.164	0.023	0.193 ± 0.023	
MSSE	SMC	0.373	0.414	0.34	0.035	0.373 ± 0.035	0.9852
	DNN-PID	0.468	0.533	0.377	0.08	0.468 ± 0.08	

Note: * indicates a statistically significant difference at $p < 0.05$, and ** indicates a statistically significant difference at $p < 0.01$.

Table 6
Indicator statistics and significance level of their superiority under different control strategies.

	Method	Mean	Max	Min	Std	mean ± SD (n = 5)	*P*-value (n = 5)
MSE	↓A	0.377	0.407	0.34	0.028	0.377 ± 0.028	
	B	3.485	4.477	3.108	0.565	3.485 ± 0.565	1.226 E-04**
	C	0.463	0.577	0.281	0.118	0.463 ± 0.118	0.09037
	D	0.709	0.821	0.599	0.086	0.709 ± 0.086	2.525 E-04**
ASE	↓A	0.193	0.215	0.164	0.023	0.193 ± 0.023	
	B	1.722	2.315	1.465	0.34	1.722 ± 0.34	2.653 E-04**
	C	0.296	0.387	0.165	0.093	0.296 ± 0.093	0.03272*
	D	0.378	0.413	0.324	0.034	0.378 ± 0.034	0.107 E-04**
MSSE	A	0.468	0.533	0.377	0.08	0.468 ± 0.08	
	B	1.899	2.805	1.524	0.514	1.899 ± 0.514	0.001516**
	C	0.454	0.548	0.318	0.088	0.454 ± 0.088	0.5965
	D	0.385	0.466	0.303	0.064	0.385 ± 0.064	0.9459

Among these, method A refers to the deviation coupling strategy employed in this study, Method B denotes equivalent control, Method C represents master–slave control, and Method D corresponds to cross-coupled adjacent control.
Note: * indicates a statistically significant difference at $p < 0.05$, and ** indicates a statistically significant difference at $p < 0.01$.

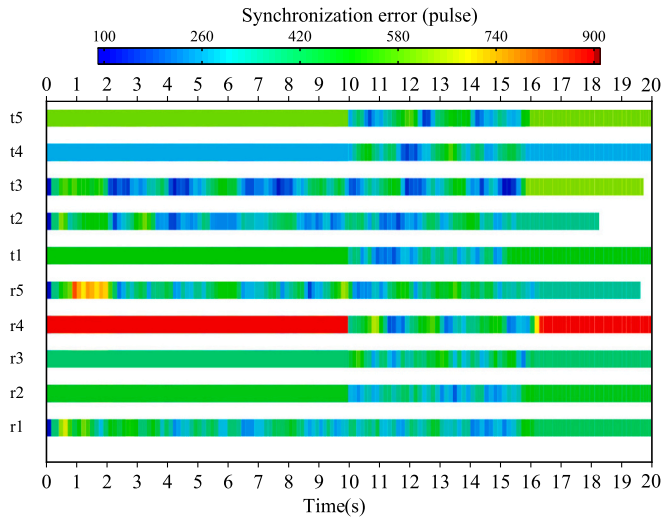


Fig. 21. Impact of transfer learning on the effectiveness of control methods.

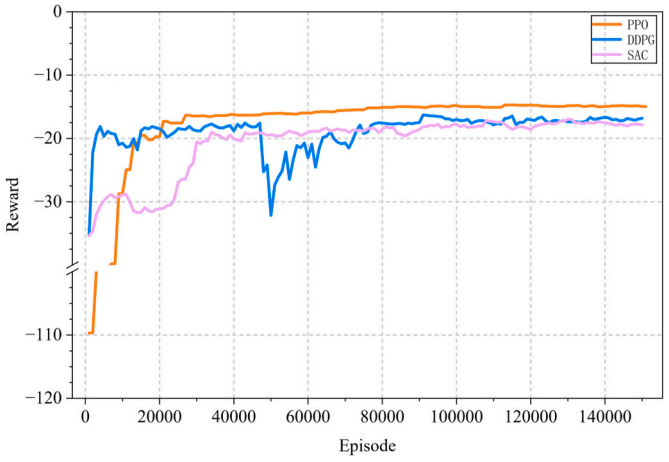


Fig. 22. Reward curves for multiple reinforcement learning algorithms.

representing a significant reduction compared to the 0.590 mm error observed without transfer learning. This is compared with the control without using transfer learning, but with random initialization parameter training. That is, the two groups of experiments in the comparison adopt different initialization methods. To ensure a structured comparison, five repeated jacking tests were conducted for each configuration. The experimental results show that the transfer learning method can significantly improve the control efficiency. Figs. r1-r5 in Fig. 21 show the performance of DNN-PID in the physical system without transfer learning. The MSEs are 0.488 mm, 0.599 mm, 0.599 mm, 0.651 mm, and 0.614 mm, respectively, with poor stability of the synchronization error and exceeding the threshold in some working conditions. Figs. t1-t5 show the fluctuation trend of the synchronization error after adding transfer learning. The MSEs are 0.355 mm, 0.377 mm, 0.451 mm, 0.437 mm, and 0.451 mm, with significant improvements in both the average and extreme value indicators of the process. The MSE decreases by an average of 0.176 mm, with a 95% confidence interval of [0.126, 0.226]. Transfer learning avoids the cumbersome process of network training and parameter tuning in the embedded device of the physical model using traditional methods, reducing the number of training sessions and achieving better control effects, saving the time cost of the training process. This result provides preliminary evidence supporting the potential future deployment of the proposed algorithm framework in physical ABM systems. It also

Table 8
Indicator statistics and significance level of their superiority under different reinforcement learning algorithms.

Indicator	Method	Mean	Max	Min	Std.	mean ± SD (n = 5)	P-value (n = 5)
MSE	↓PPO	0.046	0.057	0.032	0.011	0.046 ± 0.011	
	DDPG	0.079	0.083	0.076	0.003	0.079 ± 0.003	1.513E-03**
	SAC	0.086	0.091	0.082	0.004	0.086 ± 0.004	8.825E-04**
ASE	↓PPO	0.026	0.031	0.02	0.005	0.026 ± 0.005	
	DDPG	0.045	0.047	0.044	0.001	0.045 ± 0.001	5.720E-04**
	SAC	0.048	0.051	0.046	0.002	0.048 ± 0.002	5.199E-04**
MSSE	↓PPO	0.135	0.27	0.05	0.084	0.135 ± 0.084	
	DDPG	0.154	0.177	0.137	0.018	0.154 ± 0.018	0.3330
	SAC	0.174	0.196	0.148	0.022	0.174 ± 0.022	0.1967

Note: * indicates a statistically significant difference at $p < 0.05$, and ** indicates a statistically significant difference at $p < 0.01$.

demonstrates the advantage of integrating a digital twin monitoring infrastructure with a reinforcement learning adaptive controller, which enables real-time system observation and adaptive parameter tuning under varying load conditions. Transfer learning helps discrepancies between different systems and environments, and serves as a key mechanism that links the virtual training environment with the physical control process.

(2) Compared to other algorithms, PPO achieves at least a 10% increase in reward values. It demonstrated favorable characteristics in the four-motor synchronous control simulation system established in this study, including a faster convergence speed and better control effect. Fig. 22 shows the reward curves of the PPO algorithm, DDPG, and SAC under the same environment and training rounds. The reward curve of the PPO algorithm converges stably and tends to be stable after 40,000 training episodes, with the optimal solution reward being −14.72, which is the best among the three algorithms in this study. The convergence curves of the DDPG and SAC algorithms used for comparison show fluctuating trends at different episodes and converge after 100,000 and 120,000 episodes, respectively. The optimal solution rewards of these two algorithms during convergence are −16.45 and − 16.95, respectively, both behind the PPO algorithm. The coupled network optimized by the PPO algorithm shows a better synchronous effect throughout the process, with the synchronization error being the smallest during the entire control process. The synchronous control processes optimized using the DDPG and SAC algorithm exhibit a certain divergence in synchronization error, suggesting that their synchronization performance is inferior to that of the PPO algorithm under the same environment and reward function. A detailed comparison of the data metrics is shown in Table 8. The PPO algorithm achieves the best performance across the three metrics: MSE, ASE, and MSSE. In particular, the mean MSE decreased by 0.033 mm and 0.041 mm, with 95% confidence intervals of [0.019, 0.047] and [0.025, 0.056], respectively.

(3) Due to constraints in experimental conditions and environmental factors, several limitations remain in this study. First, the experimental validation was conducted under preset loading conditions, and the failure boundaries of the proposed control strategy have not yet been systematically explored. Extreme scenarios and

disturbances commonly encountered in real construction environments, such as actuator saturation, wind loading, structural deformation under live loads, sensor drift or failure, and communication latency, were not explicitly considered in the current experimental setup, which may influence the robustness of the proposed strategy in practical applications. Second, the physical testbed utilizes electric actuators rather than the hydraulic cylinders commonly used in full-scale ABM systems. Although this simplification facilitates controlled experiments and algorithmic validation, it cannot fully replicate the nonlinear dynamics of hydraulic actuation systems, including pressure–flow coupling, fluid compressibility, and valve hysteresis. While the scale model is configured with only 4 actuators, as opposed to the 12-cylinder hydraulic systems commonly used in real-world ABM projects. This reduction in system scale simplifies the coordination problem and reduces the complexity of actuator interactions, meaning that the synchronization performance challenges may differ. Third, although this study realized the transfer of the control strategy from simulation to a scaled model, significant differences in system complexity, uncertainty, and safety constraints exist when transferring directly to full-scale ABM equipment. In addition, the control policy is initially trained using simulation-generated data rather than real operational data, which may introduce potential discrepancies between simulated and physical conditions. However, the prior knowledge provided by transfer learning and its adaptive adjustment can partially compensate for this discrepancy. The simulation-to-scale model transfer provides a necessary intermediate step for algorithm validation and parameter adaptation. Validation on full-scale ABM systems remains necessary. Future research will therefore focus on validation on high-degree-of-freedom hydraulic multi-cylinder systems and more systematic studies of transfer learning strategies across simulation, scaled models, and full-scale ABM environments.

6. Conclusions and future work

To address the synchronous jacking problem under complex and high-risk ABM jacking conditions, this paper presented an intelligent control method for real-time synchronization error correction, together with online monitoring and feedback. Specifically, a scale model of ABM was constructed as an experimental platform. Four motors in the dynamic system of the scale model of the ABM were selected as the research objects for simulation, emulation, and transfer learning, and the existing cooperative control strategy was improved. To further enhance the construction informatization, a digital twin system was developed, enabling the visual presentation of jacking data and equipment states during synchronous control of the ABM.

In light of the experimental study, the conclusions can be summarized as follows: (1) The intelligent control algorithm based on DNN-PID has a 5.87 mm average reduction in MSE compared to the traditional PID control method. (2) Average MSE under multiple loading conditions is 0.417 mm, demonstrating good stability and adaptability. (3) Among various coupling methods, the deviation coupling method has obvious advantages, with the MSE of 0.377 mm, lower than the 3.485 mm, 0.463 mm, and 0.709 mm of other methods. (4) Transfer learning significantly reduced the synchronization error and improved the training efficiency of DNN, and has a 29% optimization effect on average MSE.

First, multiple methods including scale model experimentation, reinforcement learning, adaptive control, transfer learning, and digital twin technologies are integrated to construct an intelligent adaptive control scheme for ABM jacking control. Second, a digital twin (DT) framework is proposed that not only supports system modeling but also enables the validation and evaluation of control strategies, allowing the proposed approach to have the potential for deployment on different experimental platforms and even real ABM construction systems in future studies.

Future work could involve installing additional sensors in the scale model to monitor structural stress and strain during the jacking process. It would also be valuable to incorporate the finite element method (FEM) to assess structural safety under various operating conditions. Furthermore, integrating more operational data from actual ABMs could facilitate the transfer of control strategies to real construction scenarios. The application of physics-informed neural networks (PINN) could also be explored to embed physical constraints into the control strategy, thereby improving stability and optimization performance.

CRediT authorship contribution statement

Limao Zhang: Conceptualization, Writing – original draft, Project administration, Methodology, Investigation, Data curation, Writing – review & editing. **Xupu Zhang:** Writing – original draft, Software, Methodology, Investigation, Validation, Data curation. **Ke Chen:** Writing – review & editing, Resources. **Jiaqi Wang:** Writing – review & editing, Validation, Supervision, Methodology. **Lieyun Ding:** Writing – review & editing, Supervision, Resources.

Declaration of competing interest

We declare that the manuscript has not been submitted to, nor under review at another journal or other publishing venue. All authors have participated in (a) conception and design, analysis, and interpretation of the data; (b) drafting the article or revising it critically for important intellectual content; and (c) approval of the final version. The authors have no affiliation with any organization with a direct or indirect financial interest in the subject matter discussed in the manuscript.

Acknowledgment

The authors declare no conflict of interest. This work has been supported by the National Key Research and Development Program (Grant No. 2022YFC3802200).

Data availability

Data will be made available on request.

References

[1] C.o.T.B.a.U. Habitat, Database of Countries by Number of 150m+ Building. https://www.skyscrapercenter.com/countries?list=buildings-150, 2025.
[2] D. Wang, L. Bao, Development and prosperity of structural design of super tall buildings in China, Build. Struct. 49 (19) (2019) 11–24, https://doi.org/10.19701/j.jzjg.2019.19.003.
[3] W.D. Liu, Cause analysis and countermeasure research on construction safety accident, in: 2nd International Conference on Education, Sports, Arts and Management Engineering (ICESAME), Vol. 123, Zhengzhou, Peoples R China, 2017, pp. 273–276, 10.2991icesame-17.2017.61.
[4] X. Li, Y. Guo, F.-L. Ge, F.-Q. Yang, Human reliability assessment on building construction work at height: the case of scaffolding work, Saf. Sci. 159 (2023) 106021, https://doi.org/10.1016/j.ssci.2022.106021.
[5] J.Q. Wang, K. Chen, H. Yang, L.M. Zhang, Ensemble deep learning enabled multi-condition generative design of aerial building machine considering uncertainties, Autom. Constr. 157 (2024), https://doi.org/10.1016/j.autcon.2023.105134.
[6] L.M. Zhang, J.W. Ma, J.Q. Wang, Q. Sun, H. Yang, Multi-objective optimization for flexible design of aerial building machine under various wind conditions, Autom. Constr. 171 (2025), https://doi.org/10.1016/j.autcon.2024.105956.
[7] Z. Xia, J.Q. Wang, Y.S. Li, L.M. Zhang, C.Y. Liu, Intelligent design of key joints in aerial building machine using topology optimization and generative adversarial network, Autom. Constr. 168 (2024), https://doi.org/10.1016/j.autcon.2024.105747.
[8] J.Q. Wang, L.M. Zhang, H. Yang, H.B. Liu, M.J. Skibniewski, Dynamic reliability analysis of aerial building machine under extreme wind loads using improved QBDC-based active learning, Reliab. Eng. Syst. Saf. 244 (2024), https://doi.org/10.1016/j.ress.2024.109927.
[9] Z.B. Zuo, Y.L. Huang, X. Pan, Y.J. Zhan, L.L. Zhang, X.K. Li, M.T. Zhu, L.L. Zhang, W. De Corte, Experimental research on remote real-time monitoring of concrete strength for highrise building machine during construction, Measurement 178 (2021), https://doi.org/10.1016/j.measurement.2021.109430.

[10] J.Q. Wang, Y.Q. Fan, X. Pan, J. Sun, L.M. Zhang, Multi-source information fusion for dynamic safety risk prediction of aerial building machine using spatial-temporal multi-graph convolution network, Adv. Eng. Inform. 65 (2025), https://doi.org/10.1016/j.aei.2025.103261.
[11] Z. Zuo, X. Pan, Y. Huang, X. Wang, L. Zhang, Early warning indicators for safety monitoring and control of super high-rise ICCP, China Saf. Sci. J. 30 (1) (2020) 53–60, https://doi.org/10.16265/j.cnki.issn1003-3033.2020.01.009.
[12] L.M. Zhang, J.Q. Wang, Synchronous control of multiple hydraulic cylinders in aerial buildingmachine using improved deep reinforcement learning, Autom. Constr. 179 (2025), https://doi.org/10.1016/j.autcon.2025.106448.
[13] P.P. Liu, S.N. Song, G.F. Deng, J. Chen, Ieee, synchronous jacking control method of multi-electric cylinder based on multivariate synchronization error coupling, in: 5th International Conference on Mechatronics Technology and Intelligent Manufacturing (ICMTIM), Nanjing, Peoples R China, 2024, pp. 554–558, https://doi.org/10.1109/icmtim62047.2024.10629439.
[14] A.V. Morozov, N.A. Dobroskok, V.S. Lavrinovsiy, O.V. Mohova, IEEE, Interrelated Control of the Multi-Motor Electrical Drive, IEEE Conference of Russian Young Researchers in Electrical and Electronic Engineering (EIConRus), St Petersburg Electrotechn Univ, Russia, 2019, pp. 614–619, https://doi.org/10.1109/EIConRus.2019.8657156.
[15] F.J. Torres, G.V. Guerrero, C.D. García, J.F. Gómez, M. Adam, R.F. Escobar, Master-slave synchronization of robot manipulators driven by induction motors, IEEE Lat. Am. Trans. 14 (9) (2016) 3986–3991, https://doi.org/10.1109/tla.2016.7785923.
[16] X.Y. Zhang, H.H. Hu, H.Y. Wang, Z.Q. Wang, Overview of position synchronous control technology for multi-motor system, Syst. Sci. Control Eng. 12 (1) (2024), https://doi.org/10.1080/21642583.2024.2427074.
[17] E. Yesil, M. Guzelkaya, I. Eksin, O.A. Tekin, Cross-coupled ratio control structure, Instrum. Sci. Technol. 35 (5) (2007) 523–536, https://doi.org/10.1080/10739140701540263.
[18] F.J. Perez-Pinal, G. Calderon, I. Araujo-Vargas, IEEE, Ieee, Relative coupling strategy, IEEE International Electronic Machines and Drives Conference, Madison, Wi, 2003, pp. 1162–1166, https://doi.org/10.1109/IEMDC.2003.1210387.
[19] L. Yao, C.C. Lin, Design of gain scheduled fuzzy PID controller, in: Conference of the World-Academy-of-Science-Engineering-and-Technology, Vol. 1, London, England, 2005, pp. 152–156. https://www.researchgate.net/publication/221387932_Design_of_Gain_Scheduled_Fuzzy_PID_Controller.
[20] S.C. Tong, H.X. Li, Fuzzy adaptive sliding-mode control for MIMO nonlinear systems, IEEE Trans. Fuzzy Syst. 11 (3) (2003) 354–360, https://doi.org/10.1109/tfuzz.2003.812694.
[21] G.M. Huang, X.F. Yuan, K. Shi, X.R. Wu, A BP-PID controller-based multi-model control system for lateral stability of distributed drive electric vehicle, J. Franklin Inst. Eng. Appl. Math. 356 (13) (2019) 7290–7311, https://doi.org/10.1016/j.jfranklin.2018.12.036.
[22] J.Y. Yang, P. Zhang, X.J. Liu, Ieee, Generalized Predictive Control of PMSM, 25th Chinese Control and Decision Conference (CCDC), Guiyang, Peoples R China, 2013, pp. 816–820, https://doi.org/10.1109/CCDC.2013.6561034.
[23] Y.J. Wu, Y.B. Cheng, Y.L. Wang, Research on a multi-motor coordinated control strategy based on fuzzy ring network control, Ieee Access 8 (2020) 39375–39388, https://doi.org/10.1109/access.2020.2974906.
[24] S.R. Guendert, J.S. Houston, P.A. Wojciak, S. Cherniak, D.L. Massey, Sysplex time synchronization using IEEE 1588 Precision Time Protocol (PTP), IBM J. Res. Dev. 64 (5–6) (2020), https://doi.org/10.1147/jrd.2020.3008108.
[25] J. Zhong, Y. Chen, D. Wang, X. Liu, X. Wang, J. Wu, L. Chang, W. Chen, Industrial construction research and engineering demonstration of cast-in-place reinforced concrete high-rise buildings, Build. Sci. 38 (3) (2022) 139–145, https://doi.org/10.13614/j.cnki.11-1962/tu.2022.03.018.
[26] W.G. Jiang, L.Y. Ding, C. Zhou, Digital twin: stability analysis for tower crane hoisting safety with a scale model, Autom. Constr. 138 (2022), https://doi.org/10.1016/j.autcon.2022.104257.
[27] H. Wang, H.L. Du, Q.Y. Cui, P.F. Lui, X. Ma, A multi-motor speed synchronization control enhanced by artificial bee colony algorithm, Measur. Control 56 (1–2) (2023) 133–146, https://doi.org/10.1177/00202940221122160.
[28] B. Xie, L. Lei, Y. Yao, Y.S. Li, Advanced synchronous control of dual parallel motion systems, IEEE Trans. Ind. Electron. 70 (2) (2023) 1270–1281, https://doi.org/10.1109/tie.2022.3158018.
[29] M. Shouran, M. Alenezi, M.N. Muftah, A. Almarimi, A. Abdallah, J. Massoud, A novel AVR system utilizing fuzzy PIDF enriched by FOPD controller optimized via PSO and sand cat swarm optimization algorithms, Energies 18 (6) (2025), https://doi.org/10.3390/en18061337.
[30] W.F. Xu, S. Liu, Ieee, Synchronized Optimal Control of Intelligent Crane Car Based on Multilevel Fuzzy-PID Control, Chinese Automation Congress (CAC), Hangzhou, Peoples R China, 2019, pp. 4402–4406, https://doi.org/10.1109/cac48633.2019.8996584.
[31] S.G. Chen, F.J. Lin, C.H. Liang, C.H. Liao, Intelligent maximum power factor searching control using recurrent Chebyshev fuzzy neural network current angle controller for SynRM drive system, IEEE Trans. Power Electron. 36 (3) (2021) 3496–3511, https://doi.org/10.1109/tpel.2020.3016709.
[32] F.Z. Zhuang, Z.Y. Qi, K.Y. Duan, D.B. Xi, Y.C. Zhu, H.S. Zhu, H. Xiong, Q. He, A comprehensive survey on transfer learning, Proc. IEEE 109 (1) (2021) 43–76, https://doi.org/10.1109/jproc.2020.3004555.
[33] J.C. Kuang, G.H. Xu, T.F. Tao, Q.Q. Wu, Class-imbalance adversarial transfer learning network for cross-domain fault diagnosis with imbalanced data, IEEE Trans. Instrum. Meas. 71 (2022), https://doi.org/10.1109/tim.2021.3136175.
[34] F. Pashmforoush, M. Seyedzavvar, A transfer learning-based machine learning approach to predict mechanical properties of different material types fabricated by selective laser melting process, Proc. Inst. Mech. Eng. E 239 (3) (2025) 1400–1414, https://doi.org/10.1177/09544089231215683.
[35] T.J. Wang, Y. Tang, Transient stability preventive control based on graph convolution neural network and transfer deep reinforcement learning, Csee J. Power Energy Syst. 11 (1) (2025) 136–149, https://doi.org/10.17775/cseejpes.2022.05030.
[36] Y.F. Yin, Y. Guo, Q.R. Su, Z.T. Wang, Task allocation of multiple unmanned aerial vehicles based on deep transfer reinforcement learning, Drones 6 (8) (2022), https://doi.org/10.3390/drones6080215.
[37] D.S. Gadiraju, S.E. Azam, D. Khazanchi, SHM-traffic: DRL and transfer learning based UAV control for structural health monitoring of bridges with traffic, IeeeAccess 13 (2025) 6166–6179, https://doi.org/10.1109/access.2025.3526193.
[38] L.H. Li, B.B. Lei, C.L. Mao, Digital twin in smart manufacturing, J. Ind. Inform. Integr. 26 (2022), https://doi.org/10.1016/j.jii.2021.100289.
[39] F. Tao, B. Xiao, Q.L. Qi, J.F. Cheng, P. Ji, Digital twin modeling, J. Manuf. Syst. 64 (2022) 372–389, https://doi.org/10.1016/j.jmsy.2022.06.015.
[40] J. Viola, Y.Q. Chen, Parallel enabled and stability-aware self optimizing control with globalized constrained Nelder-Mead optimization algorithm, Ieee J. Radio Frequency Identif. 7 (2023) 178–181, https://doi.org/10.1109/jrfid.2023.3264218.
[41] L.M. Zhang, J. Guo, X.L. Fu, R.L.K. Tiong, P.H. Zhang, Digital twin enabled real-time advanced control of TBM operation using deep learning methods, Autom. Constr. 158 (2024), https://doi.org/10.1016/j.autcon.2023.105240.
[42] F. Tao, X. Ma, Q. Qi, W. Liu, H. Zhang, C. Zhang, Theory and key technologies of digital twin connection and interaction, Comput. Integr. Manuf. Syst. 29 (1) (2023) 1–10, https://doi.org/10.13196/j.cims.2023.01.001.
[43] D. Davis, M. Gronet, M. Tan, J. Thorne, Conceptual Design and Analysis of a Dynamic Scale Model of the Space Station Freedom. https://archive.org/details/NASA_NTRS_Archive_19940029730, 1994.
[44] L.M. Zhang, Y.S. Li, L.L. Wang, J.Q. Wang, H. Luo, Physics-data driven multi-objective optimization for parallel control of TBM attitude, Adv. Eng. Inform. 65 (2025), https://doi.org/10.1016/j.aei.2024.103101.